ISSN 1936-5330



Deciphering Federal Reserve Communication via Text Analysis of Alternative FOMC Statements

Taeyoung Doh, Dongho Song, and Shu-Kuei Yang
October 2020; updated October 2025
RWP 20-14
https://doi.org/10.18651/RWP2025-14



Deciphering Federal Reserve Communication via Text Analysis of Alternative FOMC Statements*

Taeyoung Doh† Dongho Song‡ Shu-Kuei Yang§

October 16, 2025

Abstract

We propose a text-based measure of monetary policy stance that models FOMC statements as convex combinations of dovish and hawkish alternatives, providing a tractable representation of the Committee's position along the policy spectrum. Leveraging staff-drafted alternative statements, we fine-tune a pre-trained language model to capture both quantitative precision and semantic tone. Stance is defined as the product of tone and novelty, and decomposed into expected and surprise components using high-frequency financial data. Surprises arise from shifts in tone relative to expectations or from statement novelty. Our framework enables counterfactuals showing how alternative communication could have moved markets.

JEL Classification: E30, E40, E50, G12.

Keywords: Alternative FOMC statements, counterfactual policy evaluation, monetary policy stance, text analysis, natural language processing.

*First version: March 2020. We thank Kyungmin Kim and Suyong Song as well as seminar participants at the Federal Reserve Bank of Kansas City, the Interfed Brown Bag, the KAEA webinar, National University of Singapore, and the University of North Carolina at Chapel Hill who provided helpful comments. All the remaining errors are ours. The author(s) thank the Center for Advancement of Data and Research in Economics (CADRE) at the Federal Reserve Bank of Kansas City (FRB-KC) for providing essential "computational/data" resources that contributed to the results reported in this paper. The opinions expressed herein are those of the authors and do not reflect the views of the Federal Reserve Bank of Kansas City or Federal Reserve System.

†Federal Reserve Bank of Kansas City; taeyoung.doh@kc.frb.org

‡Carey Business School, Johns Hopkins University; dongho.song@jhu.edu

§Federal Reserve Bank of Kansas City; shu-kuei.x.yang@kc.frb.org

Central banks increasingly turn to public communications as a tool for shaping policy actions. Public communication—particularly forward guidance—has gained prominence, especially when interest rates are constrained by the effective lower bound (e.g., Woodford, 2005; Blinder et al., 2008; elaborated upon by Bernanke, 2010). In this context, both the decisions central banks make—such as setting interest rates or adjusting asset purchases—and their explanations of these decisions are crucial to understanding how monetary policy operates.

While recent work has made progress in treating central bank statements as data, challenges remain in interpreting their textual content. Policy statements are rich in both qualitative commentary and quantitative information, blending economic assessments with explicit actions. Subtle phrasing—such as how risks are characterized or rate guidance is expressed—can shift market expectations, and its effect often depends on context. For example, "gradual rate hikes" may appear hawkish when paired with a downgraded outlook, or dovish when the outlook is upgraded. This context-sensitive nature of language means that identical phrases can carry different policy implications depending on the surrounding economic narrative—a point emphasized by Laver et al. (2003), who argue that tone must be interpreted relative to the full document.

These complexities call for natural language processing (NLP) tools capable of capturing both semantic nuance and numerical precision. We apply a context-aware pre-trained NLP model—the Universal Sentence Encoder (USE) introduced by Cer et al. (2018)—to extract the semantic content of Federal Open Market Committee (FOMC) post-meeting statements. USE generates numerical "embeddings" that represent textual meaning in a high-dimensional space. To improve its policy relevance, we fine-tune this model using synthetic examples designed to mirror the numerical information in policy actions. For instance, the model learns that a 0.50% rate hike is more hawkish than a 0.25% hike, and that 3.5% is closer to 3.75% than to 4%. This refinement allows the model to better capture how both qualitative and quantitative features contribute to policy communication.

We construct a text-based measure of monetary policy stance and decompose it into expected and surprise components using intraday financial data. We exploit a unique institutional feature: for each FOMC meeting, the staff drafts multiple versions of the policy statement—typically a more dovish "alternative A," a baseline "alternative B," and a more hawkish "alternative C" or "D." The final statement often blends language from these drafts, which span the Committee's range of views. These pre-structured alternatives provide a reference spectrum against which we position each released statement. If a statement's embedding lies closer to the dovish alternative, we classify its tone accordingly. This setup offers a transparent and adaptive measure of tone that evolves with the policy environment and internal deliberations—avoiding the rigidity of

pre-fixed tone definitions and allowing us to track shifts in what constitutes dovish or hawkish language over time.

We offer two complementary representations of monetary policy stance. From a measurement perspective, stance is defined as the product of novelty and tone—where novelty captures semantic shifts from previous statements, and tone reflects alignment with dovish or hawkish alternatives. From a structural perspective, we view the released statement as the outcome of internal deliberation, modeled as a weighted average of dovish and hawkish stances—each defined as tone times novelty. The observed stance thus reflects both the content of each position and the relative positioning of the Committee along the policy spectrum, which may reflect compromise or consensus. This interpretation echoes Ramey (2016), who notes that policy shocks can stem from shifts in the internal balance of influence. Together, these representations link textual variation not only to the conveyed policy message but also to the shifting distribution of influence within the Committee.

To isolate the surprise component of monetary policy, we begin by estimating the range of stances the market could plausibly have anticipated using a Bayesian learning framework based on pre-announcement information. We then invert observed intraday bond price movements to recover the market-expected dovish weight of the released statement—relying on the standard assumption that such movements reflect unanticipated policy news (e.g., Kuttner, 2001; Cochrane and Piazzesi, 2002; Faust et al., 2004; Gürkaynak et al., 2005). Our framework shows that monetary policy surprises arise not only from the novelty of the released statement but also from deviations in tone relative to expectations, capturing interpretable surprises that reveal shifts in the Committee's internal balance.

We validate our measure along several dimensions. First, our surprise metric aligns closely (correlations of 70–80%) with established high-frequency measures from Swanson (2017), Nakamura and Steinsson (2018), and Bauer and Swanson (2023b). Second, we find that surprise tightenings reduce stock prices and raise short-term Treasury yields, consistent with Bernanke and Kuttner (2005). Third, local projection estimates show that contractionary surprises dampen real activity and inflation while raising credit risk premia (as in Gilchrist and Zakrajšek, 2012). Finally, we show that our tone-based assumptions are plausible by cross-validating against human readings of FOMC transcripts, following guidance in Romer and Romer (2023).

Building on this foundation, our framework enables counterfactual analysis of central bank communication. Because policy stance is mapped to a unidimensional structure—the dovish weight—we simulate a spectrum of plausible alternatives by drawing from a Beta distribution centered on the realized weight. This allows greater variation when the weight is near 0.5 and less when it is near 0 or 1, reflecting uncertainty about the Committee's stance. These

counterfactuals hold market expectations fixed while varying tone and content, isolating the causal effect of language on asset prices. Conditioning on the previously estimated sensitivity of asset prices to monetary policy surprises, we translate the simulated counterfactual stances into predicted market responses. This yields confidence intervals that quantify the range of plausible financial reactions to alternative policy language. In doing so, our framework not only accounts for past market responses but also assesses the potential impact of different communication strategies.

Related Literature. Our work contributes to the growing literature that applies NLP techniques to the study of central bank communication and, more broadly, to economic and social science research. Earlier contributions have used text analysis to examine diverse contexts ranging from political discourse to macroeconomic forecasting, including Lucca and Trebbi (2009), Schonhardt-Bailey (2013), Meade and Acosta (2015), Hansen and McMahon (2016), Hansen et al. (2017), Jegadeesh and Wu (2017), Shiller (2017), Gentzkow et al. (2019), Ke et al. (2019), Shapiro and Wilson (2019), Shiller (2020), Giavazzi et al. (2020), Husted et al. (2020), Handlan (2022), Shapiro and Wilson (2021), Cieslak et al. (2022), Caldara and Iacoviello (2022), Aruoba and Drechsel (2024), Gorodnichenko et al. (2023), Schmeling and Wagner (2022), Hansen and Kazinnik (2023), Curti and Kazinnik (2023), and Shah et al. (2023).

Within this broader literature, a large number of studies apply text analysis directly to central bank communication. Much of this work relies on dictionary-based or topic-modeling techniques to quantify the tone and content of monetary policy texts—linking keywords such as "hike" or "accommodation" to categories like hawkish or dovish tone, respectively. Prominent examples include Lucca and Trebbi (2009), Hansen and McMahon (2016), Hansen et al. (2017), Husted et al. (2020), Shapiro and Wilson (2021), Acosta (2022), Cieslak et al. (2022), Aruoba and Drechsel (2024), and Schmeling and Wagner (2022). These methods offer a high degree of interpretability and are well-suited to certain tasks, such as capturing discrete shifts in tone or identifying sentiment around specific terms.

A more recent strand of the literature leverages deep learning models to better capture the semantic complexity of policy language. For example, Handlan (2022) uses XLNet to measure text-based surprises, while Gorodnichenko et al. (2023), Hansen and Kazinnik (2023), Curti and Kazinnik (2023), and Shah et al. (2023) employ BERT- or GPT-based architectures to extract latent meaning from FOMC communications and other policy texts. These models offer greater flexibility in handling nuanced language and contextual interpretation, though often at the cost of transparency or economic structure.

While the preceding studies vary in their methodological approaches, a few recent papers are especially close to ours in their economic focus. Aruoba and Drechsel (2024) apply text

analysis to internal Fed documents to approximate the central bank's information set and isolate monetary policy shocks as residuals—following the identification strategy of Romer and Romer (2004). Acosta (2022) combines high-frequency asset price data with newspaper coverage of FOMC meetings to disentangle different dimensions of policy shocks. Handlan (2022) identifies a text-based monetary policy shock as the component of intraday bond returns around FOMC announcements that is predictable from the embedding of statement language. These papers share our interest in mapping textual variation to financial market outcomes and in uncovering hidden components of policy communication.

Our study advances this line of research by introducing a rigorously structured framework that combines interpretability with analytical flexibility. We fine-tune a pre-trained NLP model (the USE; Cer et al., 2018) on synthetic FOMC-style statements with embedded numerical content, substantially enhancing its ability to extract and quantify the core quantitative elements of policy communication. Building on this, we formalize a modeling approach in which each policy statement is represented as a weighted average of dovish and hawkish stances. Our approach provides a tractable representation of internal deliberation—understood as the Committee's positioning along the policy spectrum, which may arise from consensus around a centrist stance or a balance of differing views. By defining stance as the product of tone and novelty, our framework enables a systematic decomposition of monetary policy surprises into tone- and content-driven components, offering new insights into how communication affects market reactions.[1]

Finally, we develop a novel framework for generating language-based counterfactuals grounded in central bank communication. By systematically varying the dovish weight while holding other features fixed, we simulate alternative policy stances and estimate their expected market impact. This counterfactual capability offers a transparent tool for linking internal deliberation to external financial outcomes, representing a significant advance in the empirical analysis of monetary policy communication.

The paper's structure is outlined as follows: In Section I, we introduce our NLP technique and elaborate on fine-tuning the algorithm to align with our goals. Section II provides a comprehensive explanation of the characteristics of policy alternatives and how they are crafted. Section III presents the creation of our text-based metric for monetary policy stance and outlines the identification scheme for monetary policy surprises. Section IV delves into empirical results and their policy implications. Section V examines the counterfactual effects of adopting alternative versions of the FOMC statement. Lastly, Section VI provides the conclusion.

[1] Handlan (2022) also utilizes the alternative statements in FOMC materials, interpreting the average change in expectations across them as a proxy for the Fed's private information. In contrast, we exploit the institutional structure more flexibly by allowing for time-varying influence across alternatives.

I. Universal Sentence Encoder for Text Analysis

In this section, we introduce our primary text-analysis method, the Universal Sentence Encoder (USE). USE, a deep learning-based approach, which excels in capturing semantic meaning and context within sentences, making it valuable for various NLP tasks. Additionally, we explore FinBERT (Financial Bidirectional Encoder Representations from Transformers), another powerful deep learning-based model tailored for financial text analysis. Furthermore, we introduce two widely used word counting methods: TF-IDF (Term Frequency-Inverse Document Frequency) and Latent Semantic Analysis (LSA).

While deep learning models like USE and FinBERT are adept at learning intricate textual patterns, word counting approaches like TF-IDF and LSA offer simple techniques for text analysis. Our comparative analysis aims to underscore the inherent strengths and advantages of deep learning-based models, especially in their exceptional aptitude for analyzing dense texts, such as monetary policy statements.

To ensure accessibility to general readers, we focus on highlighting the key features of various text analysis methods without delving into technical details.

A. Deep learning-based text similarity

Universal Sentence Encoder. Natural language processing tools employ a process called embedding, which converts words or texts into numerical vectors. Alternative methods of embedding differentiate diverse NLP algorithms. The USE presents two versions: 1) Deep averaging of word embeddings and 2) a Transformer-based approach utilizing the self-attention mechanism that determines the embeddings of individual words based on the semantic distance between them in a text. In the transformer-based approach, the embeddings of individual words are jointly determined based on the semantic distance between them in a text.[2] This channel captures the contextual meaning of each word within the text. These embeddings are then passed through multiple neural networks. In contrast, the deep averaging method applies multiple neural networks directly to the average of word embeddings. Despite their differences, both algorithms produce the embedding of the entire input text as their final output. For our task of calculating text similarity, we utilize the transformer-based version that captures context-dependent sentence meanings thanks to its self-attention mechanism. This capability allows it to score the similarity between texts more sensibly.

[2]The online appendix describes the transformer-based neural network architecture in detail.

For instance, consider two texts with different word lengths. The USE transforms these texts into numerical representations, obtaining two 512-dimensional vectors (U_1 and U_2) through a deep neural network architecture. After transforming texts into embedding representations, we calculate text similarity using the cosine similarity metric between vector representations of texts, denoted as U_1 and U_2:

$$\text{Sim}(\text{Text}_1, \text{Text}_2) = \text{cosine}(U_1, U_2) = \frac{U_1' U_2}{\sqrt{U_1' U_1}\sqrt{U_2' U_2}}. \tag{1}$$

For further details on the approach, please refer to Appendix A and Cer et al. (2018).

Financial Bidirectional Encoder Representations from Transformers. FinBERT, short for "Financial Bidirectional Encoder Representations from Transformers," is a specialized deep learning-based model tailored for financial text analysis. Similar to USE, it belongs to the BERT family of models, leveraging the powerful transformer architecture and producing a 768-dimensional vector embedding representation of a text. Both FinBERT and USE share the common transformer architecture, which incorporates multiple self-attention channels for efficient information processing to capture the rich pattern in the contextual meaning. By computing the cosine similarity of FinBERT embedding representations of texts, as depicted in (1), we can evaluate text similarity effectively.

Since FinBERT is trained on financial domain-specific text data, it can comprehend financial jargon, market-specific language, and economic terminologies. The key difference between FinBERT and USE lies in their training data and expertise: FinBERT is specifically trained on labeled financial texts, giving it an advantage in understanding the intricacies of the financial industry, while USE is trained on a more diverse labeled dataset covering various topics. This unique training makes FinBERT a suitable benchmark for analyzing monetary policy statements when we evaluate the performance of USE. For a more detailed explanation of the FinBERT approach, the readers can refer to Araci (2019).

B. Word counting-based text similarity

In this section, we provide a brief description of the two methods of text embedding based on word frequency counting. Gentzkow et al. (2019) provide more detailed explanations on these methods. As with the transformer-based approach, we can calculate text similarity using the cosine similarity metric of embedding representations.

Term Frequency-Inverse Document Frequency. One of the most widely used measures for text analysis is the Term Frequency-Inverse Document Frequency (TF-IDF) method. TF-IDF

calculates how important a word is in a document compared to a collection of documents. It works by looking at the frequency of words in a specific document (term frequency) and scaling it by how rarely the word appears in all the documents (inverse document frequency). This helps to prioritize words that are important in a specific document but not common across all documents.

However, TF-IDF has a significant limitation. While it is effective at identifying important words in a document, it fails to distinguish between contextual meanings created by the varying word orderings, potentially overlooking subtle nuances in the contextual meaning of words. These nuances could be crucial in interpreting monetary policy statements when a change in the word ordering may signal a shift in the priority. For example, a statement describing deteriorating labor market conditions first and then discussing inflation risk can be viewed differently from the one mentioning inflation risk first and then describing deteriorating labor market conditions.

Latent Semantic Analysis. A more sophisticated word counting method known as Latent Semantic Analysis (LSA) considers the co-frequency of words to calculate similarity scores between texts. LSA extracts a low-dimensional representation of the term-frequency/document matrix, with words and documents represented in rows and columns, respectively. This approach effectively captures common themes and underlying structures from large corpora. By rotating term frequency vectors to maximize the co-frequency of words across multiple documents, LSA extracts representations that emphasize co-occurrence patterns of words used in different texts. This makes LSA particularly effective in identifying a few key topics from a large number of texts.

Similar to the TF-IDF approach, LSA also cannot account for complex dependencies between different words beyond co-frequency, which is crucial for comprehending semantic similarity. As a result, it may not fully capture the nuanced semantic relationships between words in monetary policy statements.

C. Performance checks with simple examples

We demonstrate the superiority of the deep learning-based approaches (USE and FinBERT) in capturing contextual meaning by comparing the similarities between the following sentences:

(S_1) How old are you?

(S_2) What is your age?

(S_3) How are you?

Table 1: Similarity scores

	Deep learning-based		Word counting-based	
	USE	FinBERT	TF-IDF	LSA
Sim(How old are you, What is your age)	0.90	0.78	0.00	0.00
Sim(How old are you, How are you)	0.37	0.77	0.78	0.87

Notes: We calculate the cosine similarity scores using both deep learning-based approaches, USE and FinBERT, and word frequency counting-based approaches, TF-IDF and LSA. We apply USE version 5 for these examples.

We repeat the same exercise using word counting-based approaches (TF-IDF and LSA) for comparison.

In these examples, sentences S_1 and S_2 convey the same question, while S_3 expresses a different one. An ideal classifier should recognize that S_1 is more similar to S_2 than to S_3. Surprisingly, the similarity scores obtained using TF-IDF and LSA provide the opposite ranking, while the USE and FinBERT models produce more sensible similarity scores (see Table 1).

Word frequency counting methods, such as TF-IDF and LSA, treat words as isolated items in the dictionary and lack the ability to encode contextual linkages between words in sentences. For example, in sentences S_1 and S_3, the word "How" can have different contextual meanings, which these word counting methods fail to capture. Additionally, TF-IDF and LSA represent individual words using one-hot vectors, where only one element is non-zero, limiting their ability to comprehend semantic relationships within the text.

In contrast, deep learning models, like USE, leverage self-attention mechanisms in their neural network architectures, enabling them to understand contextual relationships between words in sentences. USE transforms word embeddings in the current layer of a neural network by computing a weighted average of all word embeddings in the text, in which weights are determined by the degree of closeness (also known as "attention") between embeddings in the previous layer. This allows USE to consider contextual linkages effectively and outperform word counting methods in tasks such as text classification and semantic similarity assessment. For instance, within the transformer architecture, the embedding representation of a word like "How" is transformed by considering the closeness between its previous layer's embedding and the embedding of the word that follows "How" in the given sentence.

D. Performance checks with policy-relevant examples

The comparison of similarity scores for the example sentences in Table 1 highlights that deep learning-based approaches perform better in capturing local context over word frequency counting-based methods. Now, we proceed to evaluate their performance on more complex sentences, especially those involving numbers, which are common in FOMC statements. Although pre-trained language models like USE and FinBERT excel at capturing semantic relationships between words and sentences, accurately representing numeric properties associated with numbers may still pose challenges for them.

To address this limitation, we undertake a fine-tuning process on the pre-trained language model using texts that contain numbers, following the approach outlined in Sundararaman et al. (2020). We choose to fine tune USE whose modest representation size can be better aligned with statements which typically consist of a few paragraphs.[3] Specifically, we create a specialized loss function that preserves the numeric distance between numbers in the sentences. In addition, we incorporate a separate loss function for datasets without numbers, ensuring that the semantic distance between sentences, as captured by the original USE representation, is maintained. This fine-tuning process enhances the model's capability to effectively handle numerical information while retaining its proficiency in understanding textual semantics.

For illustration purposes, let's consider two sentences, S_1 and S_2, which involve numbers x_1 and x_2, respectively, and two other sentences, S_3 and S_4, which do not contain numbers. The original USE representations of these sentences are denoted as U_i and U_j, where i and j represent the sentence index. In the fine-tuning step, an additional fully connected feedforward network layer is added on top of the USE representation to encode numeric information.[4]

Suppose that $f(U_i)$ and $f(U_j)$ represent the "fine-tuned" USE representation of the two sentences S_i and S_j, respectively. Training the fine-tuning layer $f(\cdot)$ involves two types of loss functions based on different pairs of sentences.

The first loss function, denoted as $\mathcal{L}_{\text{num}}(U_1, U_2)$, is constructed based on the discrepancy between the numeric distance between x_1 and x_2, and the cosine distance between $f(U_1)$ and

[3]For texts much longer than statements such as FOMC transcripts, a larger size of FinBERT can be a more suitable choice because it has a higher embedding dimension. However, our fine-turning procedure would work for both architectures. On the other hand, Compiani et al. (2025) show that USE outperforms the BERT-based sentence transformer in capturing product attributes from text data to estimate demand elasticity.

[4]A fully connected network links each element in the input layer to every element in the output layer, and the feedforward network denotes one-way connections from the input to the output layer. The online appendix describe the details of the fine-tuning procedure.

$f(U_2)$, i.e., $d\big[f(U_1), f(U_2)\big] = 1 - \text{cosine}(f(U_1), f(U_2))$:

$$\mathcal{L}_{\text{num}}(U_1, U_2) = \left(2\frac{|x_1 - x_2|}{|x_1| + |x_2|} - d\big[f(U_1), f(U_2)\big]\right)^2. \tag{2}$$

However, training the fine-tuning layer $f(\cdot)$ solely with sentences involving numbers may distort the USE representation of sentences without numbers. To address this concern, we augment the loss function $\mathcal{L}_{\text{non-num}}(U_3, U_4)$:

$$\mathcal{L}_{\text{non-num}}(U_3, U_4) = \big(d\big[U_3, U_4\big] - d\big[f(U_3), f(U_4)\big]\big)^2, \tag{3}$$

which aims to preserve the original USE representations as much as possible.

For the actual training, we create a dataset consisting of 252 sentences with numbers and 688 sentences without numbers, based on post-meeting FOMC statements. Each training data point consists of a pair of sentences, resulting in 536,848 pairs of sentences used as the training set. Among them, 63,504 pairs involve numbers, and 473,344 pairs do not. We use $\mathcal{L}_{\text{num}}(U_i, U_{-i})$ for pairs involving numbers and $\mathcal{L}_{\text{non-num}}(U_i, U_{-i})$ for pairs without numbers.

Below, we provide a few selected examples to compare the semantic distance between the original USE representations and the fine-tuned representations.

1. Policy-relevant examples without numbers
   - (P_1) Household spending has been increasing at a solid rate, on net, and business investment has been expanding;
   - (P_2) Household spending is rising moderately and business fixed investment is advancing;
   - (P_3) Household and business spending has been subdued.
2. Policy-relevant examples involving numbers
   - (N_1) FOMC decided to keep the target interest rate at 3.75 percent;
   - (N_2) FOMC decided to raise the target interest rate by 25 basis points to 4.00 percent;
   - (N_3) FOMC decided to lower the target interest rate by 25 basis points to 3.50 percent;
   - (N_4) FOMC decided to raise the target interest rate by 50 basis points to 4.25 percent.

In Panel (A) of Table 2, we present similarity scores for various pairs of (P_1, P_2, P_3) sentences. The statements P_1 and P_3 express differing perspectives on household spending, with P_1 pointing to solid growth and P_3 describing a subdued pace. On the other hand, P_2 adopts a neutral view,

Table 2: Similarity scores with policy-relevant examples

(A) Policy-relevant examples without numbers

Reference	Comparison	Similarity scores		
		Fine-tuned USE	Original USE	FinBert
(P1) has been increasing	(P2) is rising moderately	0.8032	0.8230	0.9182
	(P3) has been subdued	0.6128	0.6498	0.8662
(P2) is rising moderately	(P1) has been increasing	0.8032	0.8230	0.9182
	(P3) has been subdued	0.7060	0.7001	0.8922

(B) Policy-relevant examples involving numbers

Reference	Comparison	Similarity scores		
		Fine-tuned USE	Original USE	FinBert
(N1) keep at 3.75%	(N4) raise by 50 bps to 4.25%	0.9968	0.8831	0.9491
	(N2) raise by 25 bps to 4.00%	0.9976	0.8765	0.9412
	(N3) lower by 25 bps to 3.50%	0.9974	0.8903	0.9488
(N2) raise by 25 bps to 4.00%	(N4) raise by 50 bps to 4.25%	0.9982	0.9559	0.9891
	(N1) keep at 3.75%	0.9976	0.8765	0.9412
	(N3) lower by 25 bps to 3.50%	0.9960	0.9190	0.9880
(N4) raise by 50 bps to 4.25%	(N2) raise by 25 bps to 4.00%	0.9982	0.9559	0.9891
	(N1) keep at 3.75%	0.9968	0.8831	0.9491
	(N3) lower by 25 bps to 3.50%	0.9950	0.9133	0.9865

Notes: We compute the similarity score between the reference sentence and each of the comparison sentences. For both the original version and the fine-tuned version, USE version 5 is applied.

acknowledging a moderate increase in spending. Accordingly, the natural order of similarity scores indicates that $\text{Sim}(P_1, P_3)$ is lower than either $\text{Sim}(P_1, P_2)$ or $\text{Sim}(P_2, P_3)$.

Both the original USE representation and our fine-tuned version meet these criteria, indicating that our fine-tuning process retains the USE's proficiency in capturing semantic distance for sentences without numbers. We also observe that FinBERT satisfies these criteria, though with less distinct differentiation compared to the USE.

In Panel (B) of Table 2, we evaluate the performance of the fine-tuned USE in capturing numeric properties. Here, we provide similarity scores for different pairs of sentences (N_1, N_2, N_3, N_4), where the level of the federal funds rate varies. The similarity score should decrease as the difference in the federal funds rate levels increases. Consequently, for any pair of (N_i, N_{-i}), the lowest similarity scores should be found for (N_1, N_4), (N_2, N_3), and (N_3, N_4),

respectively. This ranking is preserved in the fine-tuned USE representation, setting it apart from the original USE and FinBERT.

The fine-tuned USE shows improved accuracy in evaluating the similarity between different statement pairs. For example, it recognizes that the phrase "keeps at 3.75%" is approximately equidistant to both "raise by 25bps to 4.00%" and "lower by 25bps to 3.50%," while the original USE considers the latter phrase much more similar due to the presence of the number 3, which is present in the original phrase. FinBERT assigns the highest similarity score to the pair "keeps at 3.75%" and "raise by 50bps to 4.25%", which contradicts key criteria for similarity assessment. This discrepancy suggests possible limitations of FinBERT in accurately capturing the nuanced semantic relationships between different statement pairs without fine-tuning to recognize numeracy when numbers are involved.

0.1 E. Discussion

The objective of these comparisons is to emphasize that even sophisticated pre-trained NLP models may have limitations in accurately analyzing central bank communications, particularly when it comes to capturing numeric properties associated with numbers. Our contribution to the literature lies in demonstrating that fine-tuning the pre-trained NLP model with an artificial dataset mimicking FOMC statements involving numbers enables a more precise assessment of crucial quantitative information, such as decisions on key interest rates, which are integral to monetary policy.

Although the variations in similarity scores for examples involving numbers are relatively small compared to the original USE algorithm, this concern does not apply when analyzing FOMC statements. The rich mixture of qualitative and quantitative descriptions in FOMC statements allows the fine-tuned USE to effectively capture semantic differences and relationships, making it a powerful tool for comprehensively analyzing and understanding monetary policy discussions.

II. Monetary Policy Alternatives

The FOMC issues official statements on interest rates, economic outlook, and policy actions. Additionally, alternative FOMC statements are created by the Federal Reserve Board staff to present various policy options. These alternatives offer a range of policy scenarios beyond the baseline path, helping policymakers assess economic risks. Policymakers can evaluate different monetary policy actions and their potential impacts, enhancing transparency in communicating future actions to markets and the public.

In this section, we provide a comprehensive explanation of the characteristics of these policy alternatives and how they are crafted as we intend to utilize information from both official and alternative policy statements. To achieve this, we draw upon the FOMC meeting presentation material dated August 9, 2011.

A. Crafting of policy alternatives

These alternative policy statements are intentionally designed to deviate from market expectations in different directions. Alternative A takes a more dovish stance, which typically involves measures such as lower interest rates or other actions aimed at stimulating economic growth and employment. On the other hand, alternative C (or D when available) adopts a more hawkish approach which often involves actions like raising interest rates or reducing monetary stimulus to combat inflation.

To gain insights into crafting alternative statements, let's explore an illustrative case. The Federal Reserve Board staff crafts these alternative statements by drawing insights from the information obtained through the Federal Reserve Bank of New York's survey of primary dealers. This survey methodically compiles data concerning market expectations for alterations in the language of the forthcoming meeting statement. As an example, here are excerpts from the August 2011 Survey of Primary Dealers in response to the question: "Do you expect any changes in the FOMC statement and, if so, what changes?"

> "Many dealers expected that the August statement would contain a downgrade to the characterization of economic conditions, and a few expected the statement to contain reference to the benchmark revisions to GDP and its impact on the outlook for economic growth. A couple of dealers expected that the statement would reference the moderation in headline inflation. The announcement of additional policy action to lengthen the duration of the SOMA portfolio was expected by a couple of dealers, as was some indication of the Committee's willingness to ease policy. Some dealers did not expect any substantial changes to the statement."
>
> (FRB of New York — Responses to Survey of Primary Dealers, August 2011)

William English, a former director of the division of monetary affairs at the Federal Reserve Board during the August 2011 meeting, cited the result of the primary dealer survey to gauge the potential market impact of the statement language:

Table 3: Overview of alternatives for the August 9, 2011 FOMC statements

	August Statement	August alternatives A	B	C
Economic activity	considerably slower than the Committee had expected	considerably slower than the Committee had expected	has been slower than the Committee had expected	had been modest of late
Target FFR	0 to $\frac{1}{4}$%	0 to $\frac{1}{4}$%	0 to $\frac{1}{4}$%	0 to $\frac{1}{4}$%
Forward guidance	exceptionally low levels at least through mid-2013	exceptionally low levels at least through mid-2013	exceptionally low levels for an extended period	exceptionally low levels for an extended period
SOMA portfolio	regularly review the size and composition of its securities holdings and is prepared to adjust those holdings as appropriate	purchase $400 billion of Treasuries with maturities 7-30 years sell an equal amount with maturities of 3 years or less... lengthening the average duration...	regularly review the size and composition of its securities holdings and is prepared to adjust those holdings as appropriate	regularly review the size and composition of its securities holdings and is prepared to adjust those holdings as appropriate
	maintain reinvestment policy	maintain reinvestment policy going forward, will use the proceeds to purchase Treasuries with remaining maturities of 7-30 years	maintain reinvestment policy	for the time being, maintain reinvestment policy

Source: Authors' construction based on **FOMC meeting presentation material, August 2011**.

> "A statement along the lines of alternative B would be about in line with the expectations captured by the Desk's survey of primary dealers last week. However, as Brian noted in his briefing, investors have become more concerned about the economic outlook in recent days and reportedly have marked up the odds associated with policy action at this meeting. Thus, the release of a statement like alternative B, with a relatively downbeat assessment of the economy and no policy action, could disappoint some market participants. Bond yields could increase and the foreign exchange value of the dollar rise. Equity prices could decline somewhat."
>
> (William English — FOMC Meeting Transcripts, August 2011)

The primary dealer survey results provide valuable insights, going beyond the market's consensus to include minority perspectives. For instance, in the August 2011 survey, a couple of

dealers anticipated additional policy actions, such as extending the duration of the SOMA portfolio. It's noteworthy that this description of extending the SOMA portfolio's duration is also found in alternative A (though absent in alternatives B or C), as detailed under the 'SOMA portfolio' section in Table 3. Conversely, the same primary dealer survey indicated that some dealers didn't anticipate significant changes to the statement. In the previous June FOMC statement, the forward guidance mentioned "exceptionally low levels...for an extended period," often interpreted as implying liftoff from the effective lower bound within a year. Notably, alternative A, as described in the 'Forward guidance' section of Table 3, modified this part of the forward guidance language to provide a more explicit time-based forward guidance. However, this change was not reflected in alternatives B or C.

The existence of various policy alternatives plays a pivotal role in stimulating discussions among FOMC members, which, in turn, promotes well-informed decision-making. For instance, there were vigorous debates involving the dovish camp, led by individuals like Chicago Fed President Charles L. Evans, advocating for the language in alternative A, and the hawkish camp, represented by figures such as Dallas Fed President Richard W. Fisher, favoring minimal language changes akin to alternative C. Despite three hawkish members dissenting from the decision, it is evident from the released official statement in Table 3 that the FOMC ultimately incorporated elements from alternative A's language, particularly by introducing a more explicit time-based forward guidance, while refraining from adopting the language of the SOMA portfolio.

The August 2011 FOMC meeting serves as a compelling illustration of how the influence of FOMC members with dovish perspectives interacts with the nuanced semantic differences between the publicly released official statement and alternative statements.

B. Policy alternatives as policy position benchmarks

Based on our illustration, it is reasonable to assume a potential alignment of market participants' views on the implications of alternative language in FOMC statements with those of the Federal Reserve Board staff.

In our paper, we contend that policy alternatives serve as valuable indicators of the market's expectations for a hawkish or dovish stance on possible policy outcomes, thereby facilitating the interpretation of the official statement's tone. This idea aligns closely with prevailing practices in text analysis, where the use of pre-labeled words or texts, such as policy alternatives in our context, is widely recognized for classifying the tone or sentiment of unlabeled text. For example, in a study by Grimmer et al. (2022), researchers explored multiple methods to determine a

politician's stance by comparing their speeches with various party platforms. We argue that the diverse perspectives on economic outlook and the associated policy recommendations found in alternative statements provide valuable reference points for interpreting the tone of the official statement. The following section formalizes our concept.

III. Text-based Monetary Policy Stance

A. Novelty and tone of released FOMC statements

We use the notation F_t to represent the FOMC statement released at time t. To quantify the novelty of each statement, we define novelty as follows:

$$\text{Novelty}_t = 1 - sim(F_t, F_{t-1}), \tag{4}$$

where $sim(F_t, F_{t-1})$ is the cosine similarity based on embeddings that capture the semantic distance between the current statement (F_t) and the previous statement (F_{t-1}) released after respective FOMC meetings. This measure allows us to assess how different or novel each FOMC statement is compared to its immediate predecessor in terms of semantic content.

Furthermore, we consider alternative FOMC statements denoted as F_t^a and F_t^c, corresponding to alternative A and C, respectively. These alternative statements are predefined with specific tones, either dovish (alternative A) or hawkish (alternative C), which enables us to evaluate the tone of the post-meeting statement. We calculate the tone using the formula:

$$\text{Tone}_t = \frac{sim(F_t^c, F_t) - sim(F_t^a, F_t)}{1 - sim(F_t^a, F_t^c)}. \tag{5}$$

Note that the tone value, as defined in (5), ranges from -1 to 1. For example, setting $F_t = F_t^a$ (or $F_t = F_t^c$) yields $\text{Tone}_t = 1$ (or -1), assuming that the semantic distance between the dovish and hawkish alternatives exceeds the distance between the released statement and either alternative. Our tone measure is consistent with the convention that a more hawkish tone is represented by a positive value, reflecting a higher implied interest rate.

B. Monetary policy stance

Our monetary policy stance combines two measures: the novelty measure (4) and the tone measure (5). Building on the concept proposed by Ke et al. (2019), we define the monetary

policy stance as:

$$\text{Stance}_t = \underbrace{(1 - sim(F_t, F_{t-1}))}_{\text{Novelty}_t} \underbrace{\left(\frac{sim(F_t^c, F_t) - sim(F_t^a, F_t)}{1 - sim(F_t^a, F_t^c)} \right)}_{\text{Tone}_t}. \tag{6}$$

An interpretation of Stance_t can be framed as follows: the semantic distance captured by our novelty term can be understood as the greatest disparity between the current meeting statement and the preceding one, while our tone measure can be seen as mitigating this disparity in a specific (toward more dovish or hawkish) direction.[5]

To clarify the interpretation of this framework, we define two reference stances that represent purely dovish and purely hawkish shifts in communication. These are given by:

$$\text{Stance}_t^{dove} = -(1 - \text{sim}(F_t^a, F_{t-1})), \quad \text{Stance}_t^{hawk} = 1 - \text{sim}(F_t^c, F_{t-1}). \tag{7}$$

Specifically, when the released statement exactly matches the dovish (or hawkish) reference—i.e., $F_t = F_t^a$ (or $F_t = F_t^c$)—the stance becomes fully dovish (or fully hawkish), such that $\text{Stance}_t = \text{Stance}_t^{dove}$ (or $\text{Stance}_t = \text{Stance}_t^{hawk}$).

C. Dovish weight in the Federal Reserve's policy communication

Dovish weight in the Federal Reserve's policy communication is defined as

$$\begin{aligned} \text{Stance}_t &= w_t \text{Stance}_t^{dove} + (1 - w_t)\text{Stance}_t^{hawk}, \\ &= 1 - sim(F_t^c, F_{t-1}) - w_t\big(2 - sim(F_t^a, F_{t-1}) - sim(F_t^c, F_{t-1})\big). \end{aligned} \tag{8}$$

By re-arranging (6) and (8), we obtain the expression for the dovish weight as follows:

$$w_t = \frac{1 - sim(F_t^c, F_{t-1}) - (1 - sim(F_t, F_{t-1}))\left(\frac{sim(F_t^c, F_t) - sim(F_t^a, F_t)}{1 - sim(F_t^a, F_t^c)} \right)}{2 - sim(F_t^a, F_{t-1}) - sim(F_t^c, F_{t-1})}. \tag{9}$$

When w_t increases (decreases), it brings the monetary policy stance of the released statement into greater alignment with the dovish (hawkish) alternative stance. This observation validates our interpretation of w_t as a dovish weight. We can readily deduce from (9) that $w_t = 1$ (0)

[5] To be precise, the distance metric between text embeddings captures not only the semantic difference but also the syntactic difference. Since the structure of FOMC statements rarely changes, the semantic dimension is likely to be dominant. But by comparing the released statement with a similar structure but different tones, we can further isolate the semantic dimension. For more details, see the online appendix.

when $F_t = F_t^a$ ($F_t = F_t^c$). Because language resembling the dovish alternative is more likely to be incorporated when dovish perspectives play a greater role in deliberation, the dovish weight reflects the extent to which those views shape the final statement. While this is generally the case, a weight near the middle does not necessarily reflect contention between camps; it may instead indicate consensus around a centrist stance.

This convex representation in (8) plays a central role in both the interpretation and application of our framework. It formalizes the idea that policy statements reflect the outcome of internal deliberation, with w_t indexing the relative positioning of the Committee along the policy spectrum. This structure echoes the perspective in Ramey (2016), where policy shocks can arise from unexpected shifts in influence among competing views. Crucially, the structure in (8) not only clarifies how the final stance reflects underlying deliberation, but also lays the foundation for counterfactual analysis—enabling us to explore how shifts in internal dynamics could have altered the communicated message and its market impact.

D. Information exchange between the market and the Fed

As detailed in Section 0.1, the Board staff formulates policy alternatives by leveraging their insights into market perceptions. Therefore, we contend that these policy alternatives serve as informative summaries of the dovish and hawkish possibilities from the market's standpoint.

Financial market participants recognize that the FOMC's policy stance is determined by deliberations among members with different policy positions. Alternative versions of FOMC statements offer a diverse array of perspectives on the spectrum of policy views held by Committee members, facilitating a reasonable estimation of alternative policy boundaries. While real-time access to the precise wording of these policy alternatives is restricted by a five-year publication delay for alternative statements, a wealth of valuable insights can still be extracted by scrutinizing meeting minutes released three weeks after the event or by analyzing intermeeting speeches.

For the sake of clarity, we use the term 'mutual information exchange' to describe the dynamic and interactive sharing of insights and information between the Federal Reserve and the market. This exchange fosters a shared understanding of economic conditions, policy intentions, and their potential consequences. In essence, we posit that it is reasonable to assume close alignment between market participants and the Federal Reserve Board staff regarding the impact of alternative language compared to the officially released statement.

E. Predicted component of monetary policy stance

Building on our perspective of the mutual information exchange between the market and the Fed, we formalize how market participants form expectations by introducing notation.

We begin by decomposing novelty into its predicted component $E_{t-\Delta}[\text{Novelty}_t]$ and its surprise component e_t:

$$\text{Novelty}_{t-\Delta} \equiv E_{t-\Delta}[\text{Novelty}_t] = \text{Novelty}_t - e_t. \tag{10}$$

Using similar notation, we designate the predicted market boundaries as $\text{Stance}^{dove}_{t-\Delta}$ and $\text{Stance}^{hawk}_{t-\Delta}$, which are established at time $t-\Delta$, immediately preceding the monetary policy announcement.

For our baseline analysis, we assume symmetric boundary conditions that differ only in sign and are set at their maximum absolute value of 1 with respect to tone:

$$\text{Stance}^{dove}_{t-\Delta} = -\text{Novelty}_{t-\Delta}, \quad \text{Stance}^{hawk}_{t-\Delta} = +\text{Novelty}_{t-\Delta}. \tag{11}$$

That is, the most dovish (hawkish) stance corresponds to the negative (positive) predicted novelty (10) just prior to the announcement.

We introduce $p_{t-\Delta}$ as the market's prevailing dovish probability, which consolidates the market's anticipated positions, encompassing both $\text{Stance}^{dove}_{t-\Delta}$ and $\text{Stance}^{hawk}_{t-\Delta}$. Based on (11), the market's expectation of monetary policy stance is given by

$$\begin{aligned} E_{t-\Delta}[\text{Stance}_t] &= p_{t-\Delta}\text{Stance}^{dove}_{t-\Delta} + (1-p_{t-\Delta})\text{Stance}^{hawk}_{t-\Delta}, \\ &= (1-2p_{t-\Delta})\text{Novelty}_{t-\Delta}. \end{aligned} \tag{12}$$

F. Monetary policy surprise

The market's response is driven by the unexpected element of the monetary policy stance, defined as

$$MPS_t \equiv \text{Stance}_t - E_{t-\Delta}[\text{Stance}_t]. \tag{13}$$

Substituting from (10) and (12), we simplify (13) to

$$\begin{aligned} MPS_t &= \text{Stance}_t - E_{t-\Delta}[\text{Stance}_t], \\ &= \underbrace{\text{Novelty}_{t-\Delta} \cdot \big(\text{Tone}_t + (2p_{t-\Delta}-1)\big)}_{\text{Tone-driven}} + \underbrace{\text{Tone}_t \cdot e_t}_{\text{Novelty-driven}}. \end{aligned} \tag{14}$$

It is essential to provide economic intuition associated with (14). The first component reflects the tone-driven surprise, holding the predicted novelty fixed. It captures how unexpected the policymaker's tone is, relative to the market's prior belief about the likelihood of a dovish versus hawkish stance, summarized by $p_{t-\Delta}$. Even if the market fully anticipates the magnitude of the novelty, a surprise can still arise if the realized tone deviates from what was probabilistically expected. The second component reflects how unexpected the magnitude of the novelty is, conditional on the chosen tone. Even when the tone aligns with prior expectations (i.e., consistent with $p_{t-\Delta}$), the surprise may still be amplified or dampened depending on how the actual novelty compares to the market's predicted novelty. This term isolates the contribution of new information, separate from interpretive differences.

G. Inferring monetary policy surprise

In the literature on identifying monetary policy shocks, intraday bond returns are frequently used as instrumental variables for monetary policy shocks, as shown in Gürkaynak et al. (2005). This approach is based on the assumption that changes in bond prices during a short time interval around policy announcements capture the immediate reaction solely to new policy information.

The relationship between intraday bond returns and monetary policy surprise is modeled as

$$r_t = -\beta MPS(p_{t-\Delta}) + \epsilon_t \tag{15}$$

where r_t is demeaned intraday bond returns; ϵ_t is white noise; $MPS(p_{t-\Delta})$, as defined in (14), is expressed (albeit with an abuse of notation) to emphasize its reliance on the probability of a dovish stance, while abstracting away from any evidence dependent on the monetary policy stance.

We search for β and $\{p_{t-\Delta}\}_{t=1}^{T}$ that maximize the rank correlation between high-frequency bond returns $\{r_t\}_{t=1}^{T}$ and the surprise component of monetary policy stance $\{MPS(p_{t-\Delta})\}_{t=1}^{T}$. In doing so, we ensure that $p_{t-\Delta}$ remains within the interval $[0, 1]$ for all t.[6] Specifically, we sort the time series of bond returns $\{r_t\}_{t=1}^{T}$ from most negative to most positive, scaling them with respect to an initial guess β. Let the ordering of the sorted-returns be indicated with new time subscripts $\{\tau_1, ..., \tau_T\}$:

$$\tilde{r}_{\tau_1} = \min\{r_t/\beta\}_{t=1}^{T}, \quad \tilde{r}_{\tau_T} = \max\{r_t/\beta\}_{t=1}^{T}. \tag{16}$$

[6] When $p_{t-\Delta}$ is not time-varying, ($p_{t-\Delta} = \overline{p}$) our estimate is identical to the maximum rank correlation estimator, see Han (1987) and Sherman (1993).

We maximize the following rank correlation function given β with respect to

$$\{p_{\tau_i-\Delta}\}_{i=1}^{T} = \text{argmax} \sum_{t \neq t'} \mathbf{1}(\tilde{r}_{\tau_t} > \tilde{r}_{\tau_{t'}}; \beta) \mathbf{1}\big(MPS(p_{\tau_t-\Delta}) < MPS(p_{\tau_{t'}-\Delta}); \beta\big). \tag{17}$$

To account for potentially multiple realizations of $\{p_{\tau_i-\Delta}\}_{i=1}^{T}$, we select the one that yields the most negative correlation by iteratively exploring β values, effectively minimizing the influence of ϵ_t in (15). Once we select $\{p_{\tau_i-\Delta}\}_{i=1}^{T}$, we can sort them back to match the original time subscript $\{p_{t-\Delta}\}_{t=1}^{T}$ and construct the corresponding $MPS(p_{t-\Delta})$ for each $t \in \{1, ..., T\}$.

We illustrate our maximum rank correlation approach with a simple example, setting $\beta = 1$ for clarity. Ignoring ϵ_t in (15), we can invert $MPS(p_{t-\Delta}) = -r_t$ and solve for $p_{t-\Delta}$ using (14). When the boundary conditions strictly hold, we can consistently recover the expected policy stance satisfying $0 < p_{t-\Delta} < 1$. However, when a boundary condition holds with equality (e.g., $\text{Stance}_{t-\Delta}^{\text{dove}} = \text{Stance}_t < \text{Stance}_{t-\Delta}^{\text{hawk}}$ and $r_t < 0$), the implied $p_{t-\Delta}$ becomes infeasible, forcing a corner solution ($p_{t-\Delta} = 1$ and $MPS = 0$). In such cases, the discrepancy must be absorbed by ϵ_t, and our measure cannot reconcile with observed returns. As emphasized by Gürkaynak et al. (2005), intraday bond returns are often used as instruments for monetary policy shocks, suggesting that ϵ_t should be minimal in most cases. The empirical frequency of such corner solutions is therefore central to evaluating the validity of our measurement strategy and underlying assumptions, which we further examine in Section 0.1.

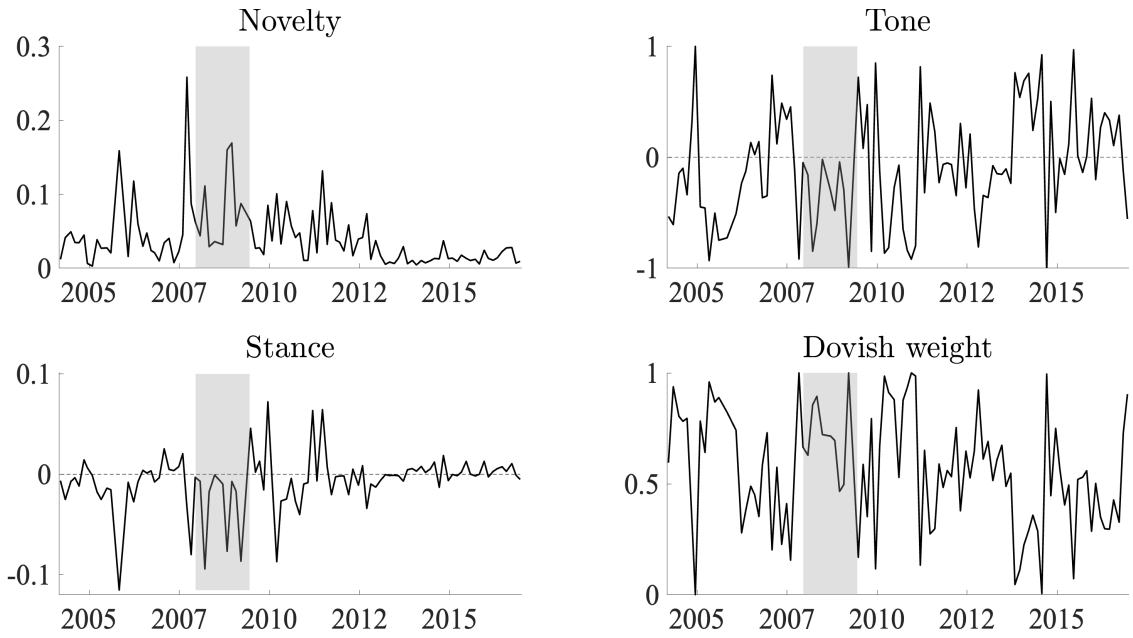
IV. Empirical Results

A. Text data

The Federal Reserve Board staff began preparing alternative FOMC statements from the March 2004 FOMC meeting. They are released to public with a five-year lag. Our dataset consists of 99 FOMC statements issued between March 2004 and December 2016. We have excluded two inter-meeting announcements (August 2007 and January 2008) and four meetings due to the unavailability of alternative statements (September 2005, December 2005, August 2008, and April 2009). In cases where multiple versions of hawkish alternative statements exist (e.g., C and D), we utilize the most extreme one (D) to identify the tone of the released statement.

During our analysis period, Lawrence H. Meyer, a former Federal Reserve Board governor (1996-2002), distributed 'monetary policy insights' newsletters to paid clients. These newsletters, issued on the last Friday before each FOMC meeting, analyzed inter-meeting macroeconomic data and FOMC member speeches to predict changes in upcoming FOMC meeting

Figure 1: Monetary policy stance and its components



Notes: We present the time-series of novelty, tone, and stance, as derived from text analysis using the definitions provided in (4), (5), and (6), respectively. Additionally, we show the dovish weight defined in (9).

statements. Starting from the September 2008 FOMC meeting, Mr. Meyer's statement drafts closely resembled the structure of actual FOMC statements. We use Mr. Meyer's statement drafts as empirical proxies for the expected FOMC statement drafts by market participants to validate our MPS measure construction assumption.

B. Text-based monetary policy measures

Figure 1 presents the time-series of novelty, tone, and stance, as derived from the USE using the definitions provided in (4), (5), and (6), respectively. Additionally, it shows the dovish weight defined in (9).

Our novelty measure demonstrates significant variability over time, allowing us to gauge the differences between the released statement and its immediate predecessor. Additionally, our tone measure spans from -1 to 1, signifying that the semantic distance between the dovish and hawkish alternative statements surpasses the disparity between the released statement and the respective alternative statements.[7] Our stance measure indicates the direction in which

[7]The sole exception occurs in September 2014, where the released statement closely resembles Alt A but is slightly more distant from Alt C than Alt A is. Alt C replaces 'considerable time after the asset purchase program ends' with 'some time after the asset purchase program ends' for forward guidance on the interest rate

the released statement differs from its immediate predecessor, either in a dovish or hawkish manner. For instance, during the Great Recession, our measure shows a significant shift in a dovish direction for the released statement. The dovish weight in the Federal Reserve's policy communication reveals that the released statement is not consistently the average of the two alternatives, as the weights can vary significantly from low to high values over time. For example, the weight for the August 2011 FOMC meeting is approximately 0.3, which reflects intense debates between the dovish and hawkish camps, as discussed in Section 0.1. The dovish camp successfully introduced explicit time-based forward guidance in the official statement. However, much of the SOMA portfolio policy descriptions did not come from the dovish alternative statements. This discrepancy is indicated by the dovish weight being 0.3. The dovish weight nearly doubled during the September 2011 FOMC meeting when the FOMC adopted a portfolio policy similar to what is described in alternative A, a feature that was missing in alternative C.

Table 4: Dovish weight and macroeconomic predictors observed before the FOMC announcement

	Correlation	p-value
Nonfarm payrolls surprise	-0.16	0.12
Employment growth	-0.13	0.20
S&P 500 change	-0.11	0.27
Yield curve slope	0.07	0.51
Commodity prices	0.09	0.38
Treasury skewness	0.05	0.66

Notes: We consider the six predictors examined in Bauer and Swanson (2023b) and present their methodology for constructing these six series here. Nonfarm payrolls surprise: the surprise component of the most recent nonfarm payrolls release prior to the FOMC announcement, measured as the difference between the released value of the statistic minus the median expectation for that release from the Money Market Services survey. Employment growth: the log change in nonfarm payroll employment from one year earlier to the most recent release before the FOMC announcement, as used in Cieslak (2018). S&P 500 change: the log change in the S&P500 stock market index from three months (65 trading days) before the FOMC announcement to the day before the FOMC announcement. Yield curve slope: the change in the slope of the yield curve from three months before the FOMC announcement to the day before the FOMC announcement, measured as the second principal component of one- to ten-year zero-coupon Treasury yields from Gürkaynak et al. (2007). Commodity prices: the log change in the Bloomberg Commodity Spot Price index (BCOMSP) from three months before the FOMC announcement to the day before the FOMC announcement. Treasury skewness: the implied skewness of the ten-year Treasury yield, measured using options on ten-year Treasury note futures with expirations in 1–3 months, averaged over the preceding month, from Bauer and Chernov (2022).

Table 4 calculates the correlation between the dovish weight and key macroeconomic predictors considered by Bauer and Swanson (2023b) which are informative about the monetary policy of the Federal Reserve. Given the range of the corresponding p-values, we narrow our focus to

path, while Alt A offers threshold-based guidance based on future inflation rate, making it challenging to classify their distance. Due to this uncertainty, we assigned the most dovish tone score to the released statement for that episode.

the three cases characterized by comparatively lower p-values. As anticipated, we discover that the dovish weight shows an inverse correlation with positive surprises in labor market indicators and the stock market return.

C. Defining expected policy boundaries

Our baseline approach constructs the predicted component of novelty, $\text{Novelty}_{t-\Delta}$, as defined in (10), to form the expected monetary policy stance—an essential input for identifying policy surprises. To this end, we estimate the following predictive regression:

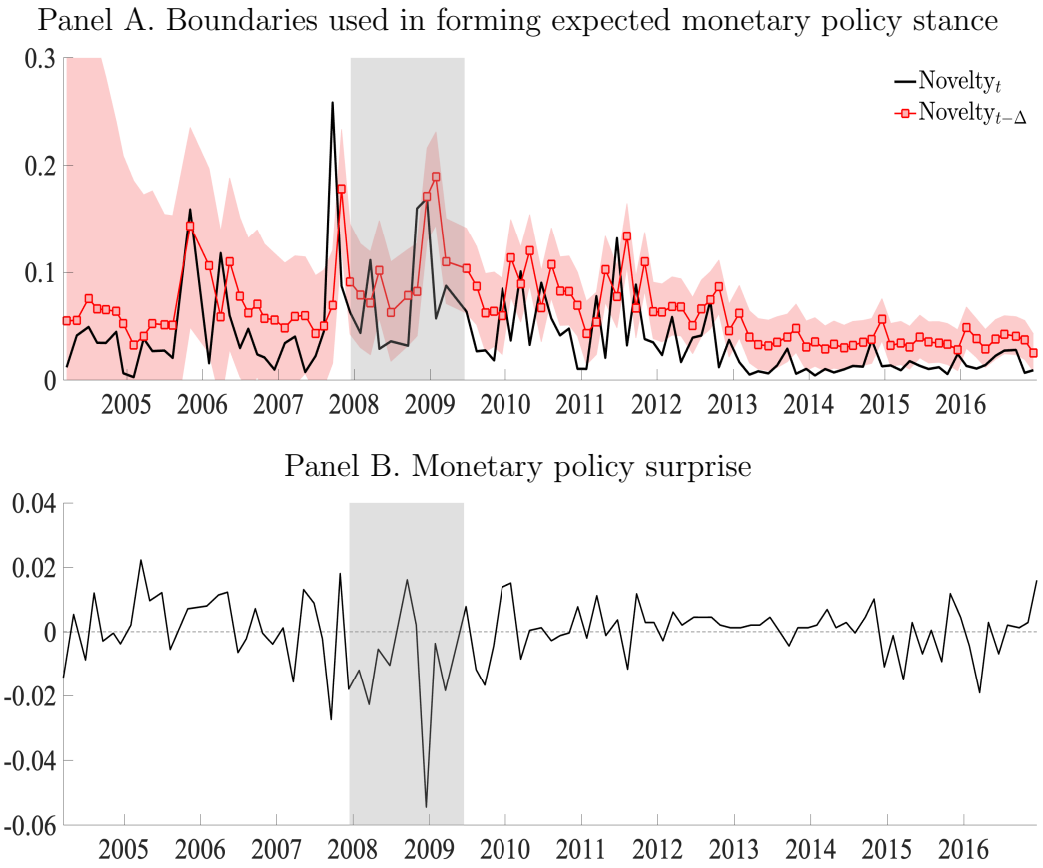
$$\text{Novelty}_t = z'_{t-\Delta}\delta + e_t, \quad e_t \sim \mathcal{N}(0, \sigma^2), \tag{18}$$

where $z_{t-\Delta}$ includes a constant term, lagged novelty, the maximum of past novelty values observed prior to the announcement, and other predictors available just before the announcement. As for the additional predictors, we leverage Mr. Meyer's draft FOMC statements, which—according to existing empirical evidence (e.g., Cieslak et al., 2018)—closely resemble the tone, structure, and policy guidance of the actual FOMC release and contain predictive information beyond publicly available data. When examining data from September 2008 to December 2016—when Meyer's draft statements are available—we observe a substantial sample correlation of approximately 0.75 between the actual novelty in the FOMC statement and the predicted novelty based on Meyer's draft released prior to the meeting. This noteworthy correlation, derived solely from Mr. Meyer's statement drafts, underscores the plausibility of assuming that market participants possess a reasonably accurate anticipation of the potential deviations or new elements that might be introduced in the forthcoming meeting statement.

To align with the market's information set, we estimate (18) in real time from the perspective of a Bayesian econometrician who, starting from an initial prior, updates beliefs about δ and σ^2 sequentially as new data arrive. At each observation t, the posterior from period t serves as the prior for period $t+1$, and so on.

We adopt an initial prior that reflects the belief that changes in the FOMC's policy statement are persistent yet allow for modest innovation. We place a Gaussian prior on δ with mean $[0.05, 0.5, 0.1, 0.4]'$. The small positive drift (0.05) reflects the belief that, on average, each statement introduces a modest amount of new information, even absent strong predictive signals. Novelty is expected to follow its past value (weight 0.5), consistent with the view that shifts in language and tone are gradual and path-dependent. A weight of 0.1 is assigned to the maximum of past novelty, reflecting the idea that market participants remain alert to the possibility of large, infrequent shifts and give some weight to the most substantial past surprise. The remaining

Figure 2: Monetary policy surprise



Notes: Top panel: The symmetric boundary conditions assumed in (11) are illustrated in this figure, based on Novelty$_{t-\Delta}$ estimated using (18). Since Novelty$_{t-\Delta}$ > Novelty$_t$ for most of the sample—reflecting our estimation specification—market participants are generally aware of the spectrum of views reflected in alternative statements. The 90% credible intervals are indicated in red. Bottom panel: We provide the time-series of monetary policy surprises (= stance − expected stance), which are defined in (14). These surprises are inferred from intraday EuroDollar futures returns with a 12-month maturity, computed over a 30-minute window around the time of announcements, following the approach of maximizing rank correlation as explained in Section 0.1.

weight of 0.4 is placed on other predictors—specifically, Meyer's draft statements—which reflect the possibility that policymakers introduce new guidance beyond prior communication.

We assume a diagonal prior variance with standard deviations set to 0.33 for all components, remaining relatively loose to let the data speak. We also impose an inverse-gamma prior $IG(10, 1)$ on the shock variance, reflecting the belief that FOMC statements generally evolve predictably, but allow for occasional, meaningful shifts. The implied prior standard deviation of about 0.33 (on a 0–1 scale) accommodates such surprises. The shape parameter of 10 corresponds to a belief that roughly 10% of each statement reflects genuinely new, unforecastable content.

We use the posterior median of the predictive regression in real time to construct $\text{Novelty}_{t-\Delta}$, illustrated in the top panel of Figure 2, and find that the results are qualitatively robust to alternative prior specifications. In most of the sample, $\text{Novelty}_{t-\Delta} > \text{Novelty}_t$, consistent with our estimation setup and the notion that market participants are generally aware of the spectrum of views represented in the alternative statements.

It is important to understand that equation (18) is not meant to reflect a literal prediction exercise conducted by market participants, but rather to mimic how they process information in practice. In forming views about upcoming FOMC statements, participants draw on past statement language (lagged novelty), the most substantial surprises in recent communication (captured by the maximum of past novelty), and informed analysis—such as Mr. Meyer's pre-meeting drafts—which distill speeches, data, and policy signals. Our regression formalizes this process in a tractable, real-time updating framework that captures how such information is plausibly absorbed and interpreted in advance of each meeting.

D. Monetary policy surprises inferred from EuroDollar futures

We utilize intraday EuroDollar futures returns with a 12-month maturity computed over a 30-minute window around the time of announcements to infer monetary policy surprises, following the approach of maximizing rank correlation as explained in Section 0.1.

As detailed in Section 0.1, we investigate instances where $p_{t-\Delta}$ approaches the boundaries of $[0, 1]$. We identify eight such cases: (i) on September 20, 2005; December 13, 2005; August 5, 2008; April 29, 2009; December 16, 2009; and March 15, 2011, where it approached 0; and (ii) on March 22, 2005 and March 18, 2008, where it approached 1. These eight instances suggest that the boundary conditions we impose may occasionally fail to yield feasible dovish probabilities. Overall, we argue that our approach reasonably captures the market's expected stance, which in the vast majority of cases (more than 90% of our sample observations) lies comfortably within the boundary defined by alternative statements.

Throughout the rest of the paper, we explore the diverse implications of monetary policy surprises derived from the baseline approach. The associated time-series is depicted in the bottom panel of Figure 2.

E. Comparison with other monetary policy surprise measures

Our surprise measure exhibits a high correlation with other measures of monetary policy shocks. Table 5 presents a sample correlation of our surprise measure with those measures.

Table 5: Comparison with other measures of monetary policy surprises

	MP surprise	MP stance
Bauer and Swanson (2023b)	0.84	0.02
- Orthogonalized measure	0.69	-0.08
Bu et al. (2020)	0.48	0.15
Nakamura and Steinsson (2018)	0.76	0.03
Swanson (2017) (FFR+FG+LSAP)	0.62	0.09
- Federal funds rate (FFR) factor	0.27	0.01
- Forward guidance (FG) factor	0.66	-0.05
- Large-scale asset purchase (LSAP) factor	-0.11	-0.21

Notes: The correlation with other existing measures of monetary policy surprises is computed using the available samples. Bauer and Swanson (2023b)'s orthogonalized measure is the residual from regressing the original Bauer and Swanson (2023b)'s monetary policy surprise measure onto six predictors in Table 4 observed before the FOMC announcement. The factors in Swanson (2017) are largely distinguished by their different loadings on the maturity spectrum of the underlying interest rate data.

Each of these approaches provides unique methods to identify monetary policy shocks, thereby contributing valuable insights into understanding the effects of monetary policy decisions on different financial assets and the overall economy. Bauer and Swanson (2023b) identify monetary policy shocks using a structural vector autoregression (VAR) model with external instruments. They use the first principal component of high-frequency changes in four Eurodollar futures rates as the instrument. Bu et al. (2020) construct monetary policy shocks based on the idea that the variance of daily bond returns is higher on FOMC days compared to non-FOMC days due to the monetary policy announcement. They use information from the entire yield curve, including near-term and long-term maturities (up to thirty years). Nakamura and Steinsson (2018) identify monetary policy shocks based on the assumption that unexpected changes in a 30-minute window during FOMC announcements arise from news about monetary policy. Swanson (2017) identifies multiple dimensions of monetary policy shocks using eight different asset prices, including three Treasury bond yields (with maturities of 2, 5, 10 years) and five interest rate futures used in Nakamura and Steinsson (2018).

Indeed, the high correlations observed between our surprise measure and those of Bauer and Swanson (2023b), Nakamura and Steinsson (2018), and Swanson (2017) indicate that our measure effectively captures the surprise component of monetary policy decisions surrounding FOMC announcements. Notably, sharing the highest correlation with Bauer and Swanson (2023b) is not unexpected, as both approaches rely on intraday EuroDollar futures returns, which can capture information about both the current federal funds rate target and the future policy path. This similarity in data sources likely contributes to the strong alignment between the two measures.

Table 6: Return predictability regression

(A) Intraday Treasury bond return prediction

Return interval	α	β	t-stat (α)	t-stat (β)	R^2
5-year maturity futures					
30 minutes	0.03	-2.00	2.08	-6.70	0.61
60 minutes	0.04	-2.22	1.81	-5.82	0.49
10-year maturity futures					
30 minutes	0.05	-2.53	1.63	-4.91	0.43
60 minutes	0.07	-2.91	1.68	-4.78	0.38

(B) Daily Treasury yield change prediction

	α	β	t-stat (α)	t-stat (β)	R^2
1-year maturity zero-coupon rate					
	-0.00	0.25	-0.46	3.13	0.26
1-year maturity instantaneous forward rate					
	-0.00	0.38	-0.05	3.39	0.31

(C) Intraday stock return prediction

Return interval	α	β	t-stat (α)	t-stat (β)	R^2
30 minutes	0.09	-1.65	1.59	-2.03	0.08
60 minutes	0.20	-3.50	3.00	-3.58	0.24

(D) Daily stock return prediction

	α	β	t-stat (α)	t-stat (β)	R^2
	0.34	-5.32	3.04	-3.39	0.21

Notes: Panel (A) examines intraday Treasury futures returns for maturities of 5- and 10-years. Panel (B) utilizes daily Treasury yield data from Gürkaynak et al. (2007). Panel (C) examines intraday E-mini S&P 500 futures returns. Panel (D) examines daily CRSP value-weighted stock returns. For panels (A) and (C), intraday returns are computed using 30- and 60-minute time intervals. To address the generated regressor's influence, t statistics are computed with bootstrap standard errors.

F. Asset return prediction with monetary policy surprises

To examine the effects of monetary policy surprises on Treasury bonds and stock prices, we conduct regressions using changes in log prices (or yields) as the dependent variable and "standardized" monetary policy surprises as the independent variable. The standardization process imposes that the daily change in the one-year maturity Treasury yield, corresponding to the

standardized monetary policy surprise, is fixed at 25 basis points.

Panel (A) of Table 6 presents the estimation results using intraday Treasury futures of maturities 5- and 10-years, respectively, computed over various time intervals for computing returns. The coefficient loading on policy surprise is consistently found to be strongly and negatively significant across different permutations, indicating a notable impact of monetary policy surprises on these Treasury futures, whose prices move inversely with Treasury yields.

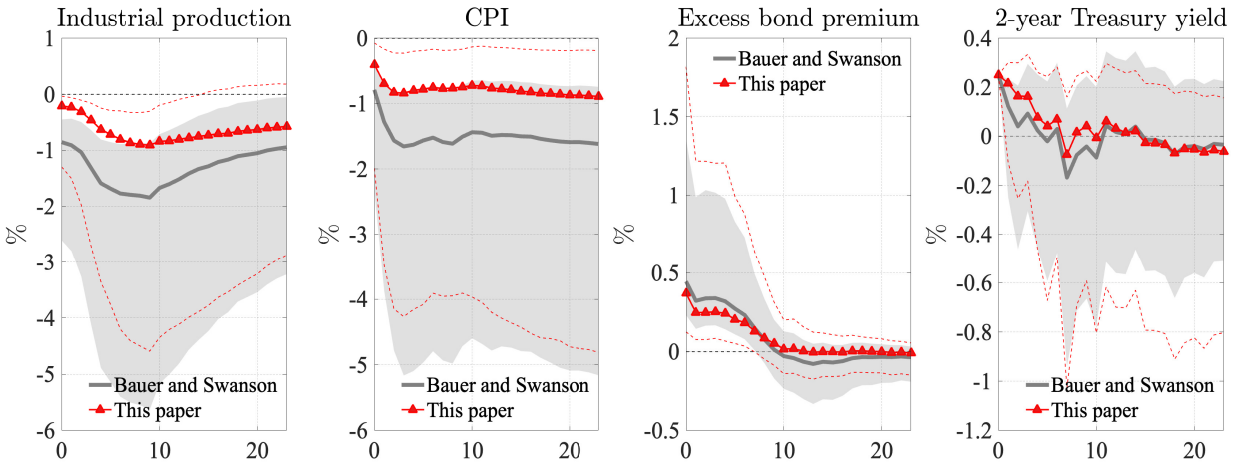
Panel (B) of Table 6 displays the estimation results from the regression involving daily one-year Treasury yield changes, considering both zero-coupon and instantaneous forward rates. A one-unit increase in monetary policy surprise is observed to significantly increase the one-year Treasury zero-coupon and forward rates by 25 and 38 basis points, respectively. The significant and consistent results from the regression analyses indicate that monetary policy surprises, as identified by the market reaction during short windows around FOMC announcements, have lasting impacts on short-term Treasury yields beyond narrow windows, moving in the intended direction.

Given that the Federal Reserve intervenes in Treasury markets to influence interest rates, it is not surprising that bonds react to monetary policy surprises. However, the impact of monetary policy extends beyond bond returns and influences the broader financial market conditions, including the stock market. Understanding the connections between monetary policy and asset prices, beyond bond returns, is crucial for evaluating monetary policy transmission channels, as emphasized by Bernanke and Kuttner (2005).

The estimation results, summarized in Panel (C) and (D) of Table 6, indicate that monetary policy surprise significantly predicts stock returns at various window intervals. On average, a 25 basis point increase in one-year Treasury yields leads to a 2–3 percentage point decline in stock prices. The R^2 values are approximately 20% across different return window intervals. Our findings are consistent with Bernanke and Kuttner (2005) in showing that monetary tightening significantly lowers stock returns.

Readers familiar with Bernanke and Kuttner (2005) may perceive our estimated stock return response as relatively large. However, this reflects differences in return frequency and estimation sample: Bernanke and Kuttner (2005) rely on daily returns, whereas our analysis focuses on intraday responses. Moreover, when we use high-frequency monetary policy shocks from Bauer and Swanson (2023b) and Nakamura and Steinsson (2018)—each normalized to generate a 25 basis point increase in the one-year Treasury yield, as in our baseline and using the same estimation sample—we obtain slope estimates that are comparable to, or larger than, our own. This suggests that our estimate lies at the lower end of the range implied by more recent high-frequency approaches.

Figure 3: SVAR with high-frequency monetary policy surprise as an external instrument



Notes: Following Bauer and Swanson (2023b), we estimate a SVAR with a 12-month lag using the following variables: log industrial production, log consumer price index, excess bond premium Gilchrist and Zakrajšek (2012), and the two-year Treasury bond yield. To instrument our analysis, we employ the orthogonalized monetary policy surprise measure derived from the residuals of a regression on six predictors explained in Table 4 observed before the FOMC announcement. While we estimate the VAR coefficients using a longer sample from 1973:M1 to 2019:M12, the orthogonalized high-frequency monetary policy surprises are available only from 2004:M3 to 2016:M12. To facilitate comparison, we present the impulse responses obtained by utilizing the orthogonalized high-frequency monetary policy surprise instrument employed in Bauer and Swanson (2023b).

G. Macroeconomic impacts of monetary policy surprises

In this section, we employ a similar approach to Bauer and Swanson (2023b) to assess the effects of a monetary policy shock on various macroeconomic variables. Specifically, we investigate the effects of a monetary policy shock on log industrial production, log consumer price index, the excess bond premium by Gilchrist and Zakrajšek (2012), and the two-year Treasury bond yield. We use our orthogonalized monetary policy surprise measure as an instrument z_t for the four-variable monthly VAR, with a lag length of 12 represented by[8]

$$y_t = A_0 + \sum_{k=1}^{12} A_k y_{t_k} + u_t, \quad u_t = S\epsilon_t \sim (0, \Omega), \quad \epsilon_t \sim (0, I_4). \tag{19}$$

Here, y_t denotes the vector of macroeconomic variables, and A_0, A_k, and u_t are the coefficient matrices and structural residuals, respectively. The matrix S contains the identified structural shocks ϵ_t from reduced-form VAR residuals u_t.

We standardize a one-standard deviation positive monetary policy shock to raise the two-year Treasury bond yield by 25 basis points. Subsequently, we estimate the impulse responses of the

[8]The orthogonalization procedure for the monetary policy surprise is detailed in the legend of Table 5.

vector of macroeconomic variables y_t to this monetary policy shock $\epsilon_{mp,t}$. The VAR coefficients are inferred from an extended sample covering the period from 1973:M1 to 2019:M12, while the column in S that identifies a monetary policy shock is estimated using orthogonalized high-frequency monetary policy surprises observed between 2004:M3 and 2016:M12. To facilitate comparison, we re-estimate the SVAR using Bauer and Swanson (2023b)'s instrument while matching the same sample availability and overlay the results. The respective impulse response functions of the VAR variables are displayed in Figure 3.[9]

Given the high correlation between our surprise measure and that of Bauer and Swanson (2023b) (as shown in Table 5), it is not surprising that the two sets of impulse responses exhibit similarities. However, due to the relatively short estimation sample from 2004 to 2016, the confidence intervals are wide, which may impact the statistical significance of the impulse responses. This observation is reflected in the uncertainty surrounding the estimates, particularly in the short-term.

In summary, our findings suggest that our text-based surprise measure, being highly correlated with Bauer and Swanson (2023b)'s measure, demonstrates similar impulse responses of macroeconomic variables to a monetary policy shock, thus further supporting the effectiveness of our approach in capturing the impact of monetary policy decisions.

V. Counterfactual Policy Evaluation

The counterfactual monetary policy stance represents the stance that would have been conveyed had an alternative statement—reflecting a different internal deliberation outcome—been released. Consistent with the deliberation-based structure in (8), we express the counterfactual stance as a convex combination of the dovish and hawkish stances, indexed by the counterfactual dovish probability w_t^{CF}:

$$\text{Stance}_t^{CF} \mid w_t^{CF} = w_t^{CF} \cdot \text{Stance}_t^{dove} + (1 - w_t^{CF}) \cdot \text{Stance}_t^{hawk}. \tag{20}$$

To capture counterfactual scenarios that reflect reasonable variation in policy deliberation, we consider hypothetical outcomes in which the FOMC's communication could have differed slightly from the released statement. These scenarios represent alternative conclusions the Committee might have reached during the meeting. We consider these to be "modest policy interventions"

[9] Jentsch and Lunsford (2022) show that the wild bootstrap used in the SVAR of Bauer and Swanson (2023a) ignores the dependency among observations and propose a residual-based moving block bootstrap (MBB). We check the robustness of our confidence intervals using this method in the appendix.

in the sense of Leeper and Zha (2003). Accordingly, we model the counterfactual dovish weight w_t^{CF} to be centered at the realized dovish weight w_t, as defined in (9), allowing for moderate deviations:

$$E[w_t^{CF} \mid w_t] = w_t, \quad \text{Var}[w_t^{CF} \mid w_t] = \zeta\, w_t(1 - w_t), \tag{21}$$

where $\zeta \in (0, 1)$ is a dispersion parameter. Intuitively, the conditional variance of the counterfactual stance is highest when the distribution is centered near 0.5, reflecting a wider range of plausible alternative statements. The variance declines as the distribution becomes more concentrated near 0 or 1, where the stance is more clearly aligned with one end of the policy spectrum. Given this conditional mean and variance, the counterfactual weight w_t^{CF} follows a Beta distribution:

$$w_t^{CF} \sim \text{Beta}\left(w_t\left(\frac{1-\zeta}{\zeta}\right), (1 - w_t)\left(\frac{1-\zeta}{\zeta}\right)\right). \tag{22}$$

Given a draw of the counterfactual dovish weight, we compute the change in the monetary policy surprise resulting from the counterfactual experiment as

$$MPS_t^{CF} \mid w_t^{CF} - MPS_t = w_t^{CF} \cdot \text{Stance}_t^{dove} + (1 - w_t^{CF}) \cdot \text{Stance}_t^{hawk} - \text{Stance}_t, \tag{23}$$

since the expected stance, $E_{t-\Delta}[\text{Stance}_t]$, is held constant and cancels out in the difference.

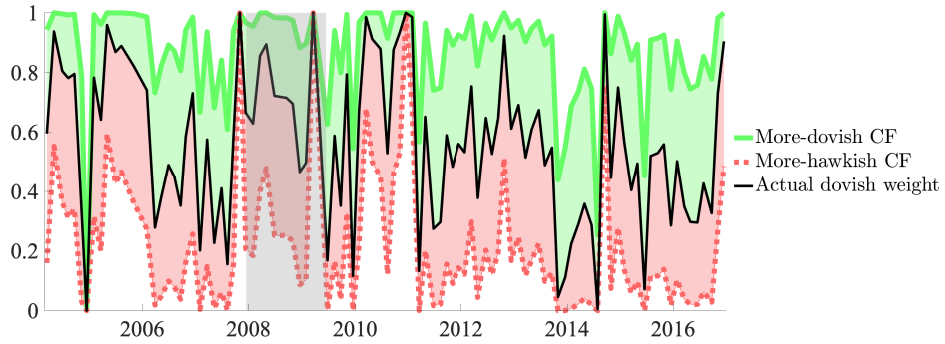
To assess the asset market impact of a counterfactual policy announcement, we condition on the estimated slope coefficient $\hat{\beta}$ from the test asset returns (stocks, in our context) and a given counterfactual weight w_t^{CF}:

$$\widehat{r}_t^{CF} \mid (\hat{\beta}, w_t^{CF}) = \hat{\beta} \cdot \left(w_t^{CF} \cdot \text{Stance}_t^{dove} + (1 - w_t^{CF}) \cdot \text{Stance}_t^{hawk} - \text{Stance}_t\right). \tag{24}$$

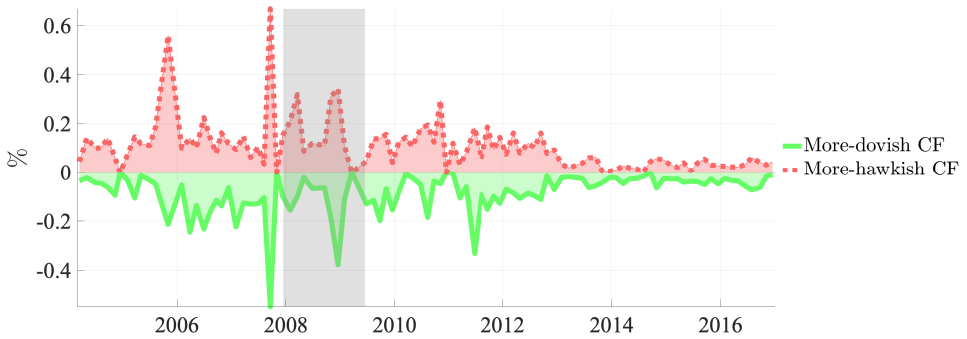
We simulate w_t^{CF} repeatedly from the distribution in (22) to generate a distribution of counterfactual returns implied by (24), effectively capturing the range of asset price responses to alternative statement formulations (see Figure 4). Relative to the baseline dovish weight (indicated by the black line), we mark regions where $w^{CF} > w$ with green shading and regions where $w^{CF} \leq w$ with red shading. As implied by (22), the dispersion of counterfactual weights is smallest when the baseline dovish weight is near 0 or 1 and largest when it is near 0.5. For this figure, we use $\zeta = 0.2$, which we view as a reasonable value: it yields a counterfactual distribution that is neither too narrow nor too diffuse and adequately captures plausible variations without pushing weight mass toward the extremes.

Figure 4: Asset responses to counterfactual policy scenarios

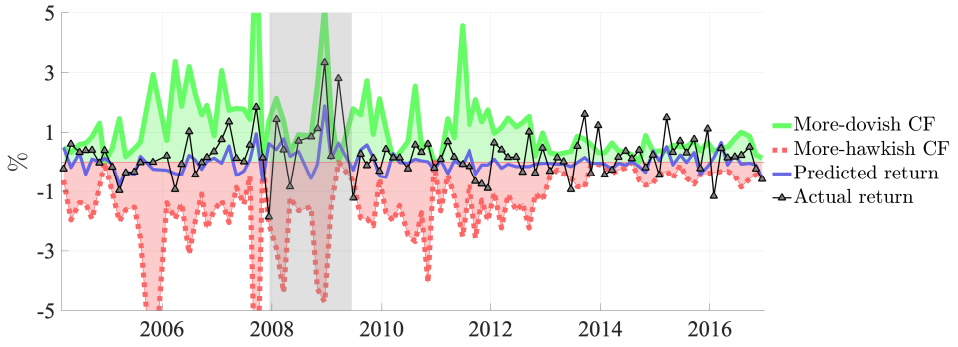
Panel A. Distribution of counterfactual dovish weights



Panel B. Distribution of Treasury yield responses to counterfactual policy



Panel C. Distribution of stock return responses to counterfactual policy



*Notes:*In the top panel, the shaded areas represent the 90% interval of simulated dovish weights based on $\zeta = 0.2$ in (22). The green-shaded region marks simulations where $w^{CF} > w$, while the red-shaded region marks those where $w^{CF} \leq 0$. The black solid line indicates the conditional mean of the simulations, which, by construction, matches the baseline dovish weight. In the middle and bottom panels, the shaded areas depict the 90% interval of simulated Treasury yield and stock return responses to the counterfactual scenarios in (24), again split into green and red based on whether the counterfactual stance is more or less dovish than realized. Gray bars indicate NBER recession periods.

This simulation-based framework allows us to assess whether such deviations lead to economically significant changes in financial markets and to quantify the associated uncertainty. We

summarize the return distributions using simulated quantile bands, which reflect uncertainty arising from alternative—but plausible—shifts in communication tone. These intervals provide a basis for evaluating whether deviations from the released statement could have led to meaningful differences in asset prices. Since higher dovish weights imply more accommodative policy stances, counterfactuals with $w^{CF} > w$ tend to generate higher stock returns (green-shaded), while those with $w^{CF} \leq w$ yield lower returns (red-shaded).

To put these movements in perspective, a 10 basis point decline (increase) in the one-year Treasury yield corresponds to roughly a 1.4% increase (decrease) in intraday stock returns over a one-hour window. This elasticity is implied by our regression estimate reported in Table 6, where $\hat{\beta} = -3.5$ under a 60-minute return interval—the specification that yields the highest R^2. As noted in the previous section, this response is consistent with those implied by high-frequency monetary policy shocks from Bauer and Swanson (2023b) and Nakamura and Steinsson (2018), and lies at the lower end of estimates based on comparable data. We therefore view the resulting counterfactual return magnitudes as empirically reasonable.

For reference, we also plot the predicted return values (purple line) implied by the monetary policy surprise from our baseline analysis, given by $\hat{r} = \hat{\beta} \cdot \text{MPS}$. In terms of magnitude, about 60% of counterfactual return responses across the full time series fall within ±1%, and approximately 85% fall within ±2%. To interpret these values in yield-equivalent terms, we use our estimated stock-yield elasticity of 1.4% per 10 basis point move in the one-year Treasury yield. This implies that ±1% (±2%) return responses correspond to yield movements of approximately ±7 (±14) basis points.

As a case study, we revisit the August 2011 FOMC meeting, for which our baseline analysis assigns a dovish weight of 0.3 to the released statement. Our counterfactual exercise quantifies the potential market impact of more hawkish alternatives that were under active consideration at the time. Specifically, reducing the dovish weight to 0.2 leads to an estimated 27 basis point decline in intraday stock returns; a further reduction to 0.1 implies a 55 basis point decline. These estimates give empirical substance to concerns raised by English and others during the deliberation (see Section 0.1), where more hawkish language was seriously debated. Releasing a more dovish version of the statement, by contrast, would have produced gains of similar magnitude. This exercise highlights the market consequences of marginal shifts in communication tone, even within a narrow set of plausible policy alternatives.

To ensure that our counterfactual exercises remain grounded in plausible variation, we do not assume deterministic or wholesale replacements of policy statements. Instead, the framework permits continuous, probabilistic deviations in the policy stance. By drawing counterfactual dovish weights from a Beta distribution centered on the realized weight, we capture a spectrum

of alternative—but reasonable—conclusions that the Committee might have reached in the course of deliberation. Incrementally adopting language from alternative statements—while not inherently unrealistic—does not guarantee monotonic variation in stance, as the mapping from language to policy content is complex and nonlinear.[10]

A caveat of our approach is that it conditions on a fixed slope coefficient $\hat{\beta}$, which is estimated using the full sample. By holding $\hat{\beta}$ fixed, we abstract from uncertainty in the return–policy relationship. In principle, one could allow $\hat{\beta}$ to evolve over time to reflect changing sensitivities of asset prices to monetary policy surprises. While this limitation is unlikely to materially affect the interpretation of our counterfactual exercises, it may be more relevant in real-time policy evaluation.

VI. Conclusion

This paper develops a text-based framework for quantifying the stance of monetary policy communication and assessing its effects on financial markets. We model FOMC post-meeting statements as weighted averages of dovish and hawkish alternatives—reflecting internal deliberation along the policy spectrum—and define stance as the product of tone and novelty. To extract this structure, we fine-tune the Universal Sentence Encoder (USE; Cer et al., 2018) on synthetic FOMC-style statements with embedded numerical content, enhancing the model's ability to capture both semantic nuance and quantitative precision in policy language.

Combining this enhanced text analysis with high-frequency bond market data, we construct a measure of monetary policy surprises that decomposes announcements into tone and novelty components. We find that surprise tightenings lower equity prices, while surprise easings raise them—consistent with asset pricing intuition and earlier studies (e.g., Bernanke and Kuttner, 2005; Swanson, 2017; Nakamura and Steinsson, 2018). Our stance model allows us to interpret these surprises as shifts in the Committee's internal balance, linking textual variation directly to market responses.

Building on this structure, we conduct counterfactual simulations by varying the dovish weight to generate alternative policy stances. These counterfactuals are grounded in the institutional structure of FOMC communication—specifically, the staff-drafted alternative A/B/C statements—and avoid unrealistic substitutions of full texts by instead allowing for modest, probabilistic deviations consistent with deliberation (see Leeper and Zha, 2003). The simulations

[10] A generative artificial intelligence model using USE may generate identical text corresponding to multiple values of our counterfactual weights, but we leave this to future research.

reveal that even subtle shifts in tone, within the feasible range of language under consideration, can generate economically meaningful movements in asset prices.

Our findings underscore the role of central bank statements as active instruments of policy and demonstrate how language-based tools can be used to assess their effects rigorously. As large-scale language models continue to improve human-level understanding and reasoning in text (see Manning, 2022), our study illustrates the promise of integrating such models into economic analysis. While our primary application is to post-meeting FOMC statements, the approach of combining fine-tuned language models with structural modeling of policy deliberation has broader potential—for instance, in analyzing forward guidance, speeches, or other forms of narrative communication (see Shiller, 2017; Shiller, 2020).

References

Acosta, Miguel, "A New Measure of Central Bank Transparency and Implications for the Effectiveness of Monetary Policy," 2022. Manuscript.

Araci, Dogu, "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models," 2019.

Aruoba, Boragan and Thomas Drechsel, "Identifying Monetary Policy Shocks: A Natural Language Approach," *American Economic Journal: Macroeconomics*, 2024. forthcoming.

Bauer, Michael D. and Eric T. Swanson, "An Alternative Explanation for the 'Fed Information Effect'," *American Economic Review*, March 2023, *113* (3), 664–700.

_ and _, "A Reassessment of Monetary Policy Surprises and High-Frequency Identification," *NBER Macroeconomics Annual*, 2023, *37* (1), 87–155.

_ and Mikhail Chernov, "Interest Rate Skewness and Biased Beliefs," *Journal of Finance*, 2022.

Bernanke, Ben S., "Central Bank Independence, Transparency, and Accountability," in "Speech of Chairman Ben S. Bernanke at the Institute for Monetary and Economic Studies International Conference" 2010.

Bernanke, Ben S and Kenneth N Kuttner, "What explains the stock market's reaction to Federal Reserve policy?," *The Journal of finance*, 2005, *60* (3), 1221–1257.

Blinder, Alan S., Michael Ehrmann, Marcel Fratzscher, Jakob De Haan, and David-Jan Jansen, "Central Bank Communication and Monetary Policy: A Survey of Theory and Evidence," *Journal of Economic Literature*, 2008, *46* (4), 910–945.

Bu, Chunya, John Rogers, and Wenbin Wu, "Forward-Looking Monetary Policy and the Transmission of Conventional Monetary Policy Shocks," 2020.

Caldara, Dario and Matteo Iacoviello, "Measuring geopolitical risk," *American Economic Review*, 2022, *112* (4), 1194–1225.

Cer, Daniel, Yinfei Yang, Shengyi Kong, Nan Hua, Nicole Limtiaco, Rhomni St. John, Noah Constant, Mario Guajardo-Cespedes, Steve Yuan, Chris Tar, Yun-Hsuan Sung, Brian Strope, and Ray Kurzweil, "Universal Sentence Encoder," 2018. Manuscript.

Cieslak, Anna, "Short-Rate Expectations and Unexpected Returns in Treasury Bonds," *The Review of Financial Studies*, 04 2018, *31* (9), 3265–3306.

_ , Adair Morse, and Annette Vissing-Jorgensen, "Stock Returns over the FOMC Cycle," *Journal of Finance*, 2018, *74* (5), 2201–2248.

_ , Stephen Hansen, Michael McMahon, and Song Xiao, "Policymakers' Uncertainty," *Manuscript*, 2022.

Cochrane, John H. and Monika Piazzesi, "The Fed and Interest Rates - A High-Frequency Identification," *American Economic Review*, May 2002, *92* (2), 90–95.

Compiani, Giovanni, Ilya Morozov, and Stephan Seiler, "Demand estimation with text and image data," *arXiv preprint arXiv:2503.20711*, 2025.

Curti, Filippo and Sophia Kazinnik, "Let's face it: Quantifying the impact of nonverbal communication in FOMC press conferences," *Journal of Monetary Economics*, 2023.

Faust, Jon, Eric T. Swanson, and Jonathan H. Wright, "Identifying VARS based on high frequency futures data," *Journal of Monetary Economics*, 2004, *51* (6), 1107–1131.

Gentzkow, Matthew, Bryan Kelly, and Matt Taddy, "Text as Data," *Journal of Economic Literature*, 2019, *57* (3), 535–574.

Giavazzi, Francesco, Felix Iglhaut, Giacomo Lemoli, and Gaia Rubera, "Terrorist Attacks, Cultural Incidents and the Vote for Radical Parties: Analyzing Text from Twitter," Technical Report, National Bureau of Economic Research 2020.

Gilchrist, Simon and Egon Zakrajšek, "Credit Spreads and Business Cycle Fluctuations," *American Economic Review*, June 2012, *102* (4), 1692–1720.

Gorodnichenko, Yuriy, Tho Pham, and Oleksandr Talavera, "The voice of monetary policy," *American Economic Review*, 2023, *113* (2), 548–584.

Grimmer, Justin, Margaret E Roberts, and Brandon M Stewart, *Text as data: A new framework for machine learning and the social sciences*, Princeton University Press, 2022.

Gürkaynak, Refet S, Brian Sack, and Eric Swanson, "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements," *International Journal of Central Banking*, 2005, *1*, 55–93.

_ , _ , and Jonathan Wright, "The US Treasury yield curve: 1961 to the present," *Journal of Monetary Economics*, 2007, *54* (8), 2291–2304.

Han, Aaron K., "Non-parametric Analysis of a Generalized Regression Model," *Journal of Econometrics*, 1987, *35*, 303–316.

Handlan, Amy, "Text Shocks and Monetary Surprises: Text Analysis of FOMC Statements with Machine Learning," 2022. Manuscript.

Hansen, Anne Lundgaard and Sophia Kazinnik, "Can ChatGPT Decipher Fedspeak?," 2023. Manuscript.

Hansen, Stephen and Michael McMahon, "Shocking language: Understanding the macroeconomic effects of central bank communication," *Journal of International Economics*, 2016, *99*, S114–S133. 38th Annual NBER International Seminar on Macroeconomics.

_ , _ , and Andrea Prat, "Transparency and Deliberation Within the FOMC: A Computational Linguistics Approach," *Quarterly Journal of Economics*, 2017, *133* (2), 801–870.

Husted, Lucas, John Rogers, and Bo Sun, "Monetary policy uncertainty," *Journal of Monetary Economics*, 2020, *115*, 20–36.

Jegadeesh, Narasimhan and Di Andrew Wu, "Deciphering Fedspeak: The information content of FOMC meetings," *Available at SSRN 2939937*, 2017.

Jentsch, Carsten and Kurt G Lunsford, "Asymptotically valid bootstrap inference for proxy SVARs," *Journal of Business & Economic Statistics*, 2022, *40* (4), 1876–1891.

Ke, Zheng Tracy, Bryan T. Kelly, and Dacheng Xiu, "Predicting Returns with Text Data," 2019. Manuscript.

Kuttner, Kenneth N, "Monetary policy surprises and interest rates: Evidence from the Fed funds futures market," *Journal of Monetary Economics*, 2001, *47* (3), 523–544.

Laver, Michael, Kenneth Benoit, and John Garry, "Extracting Policy Positions from Political Texts Using Words as Data," *American Political Science Review*, 2003, *97* (2), 311–331.

Leeper, Eric M. and Tao Zha, "Modest policy interventions," *Journal of Monetary Economics*, 2003, *50* (8), 1673–1700.

Lucca, David O and Francesco Trebbi, "Measuring central bank communication: an automated approach with application to FOMC statements," Technical Report, National Bureau of Economic Research 2009.

Manning, Christopher D, "Human language understanding & reasoning," *Daedalus*, 2022, *151* (2), 127–138.

Meade, Ellen E. and Miguel Acosta, "Hanging on Every Word: Semantic Analysis of the FOMC's Postmeeting Statement," FEDS Notes, Board of Governors of the Federal Reserve System (U.S.) 2015.

Nakamura, Emi and Jón Steinsson, "High-frequency identification of monetary non-neutrality: the information effect," *The Quarterly Journal of Economics*, 2018, *133* (3), 1283–1330.

Ramey, Valerie A, "Macroeconomic shocks and their propagation," *Handbook of macroeconomics*, 2016, *2*, 71–162.

Romer, Christina D. and David H. Romer, "A New Measure of Monetary Shocks: Derivation and Implications," *American Economic Review*, September 2004, *94* (4), 1055–1084.

_ and _, "Presidential Address: Does Monetary Policy Matter? The Narrative Approach after 35 Years," *American Economic Review*, June 2023, *113* (6), 1395–1423.

Schmeling, Maik and Christian Wagner, "Does Central Bank Tone Move Asset Prices?," *Available at SSRN 2629978*, 2022.

Schonhardt-Bailey, Cheryl, *Deliberating American Monetary Policy: A Textual Analysis*, MIT Press, 2013.

Shah, Agam, Suvan Paturi, and Sudheer Chava, "Trillion Dollar Words: A New Financial Dataset, Task & Market Analysis," 2023.

Shapiro, Adam and Daniel J Wilson, "The Evolution of the FOMC's Explicit Inflation Target," *Federal Reserve Bank of San Francisco Working Paper 2019-02*, 2019, *2019*, 12.

_ **and** _ , "Taking the Fed at its Word: A New Approach to Estimating Central Bank Objectives using Text Analysis," *The Review of Economic Studies*, 12 2021, *89* (5), 2768–2805.

Sherman, Robert P., "The Limiting Distribution of the Maximum Rank Correlation Estimator," *Econometrica*, 1993, *61* (1), 1123–137.

Shiller, Robert J, "Narrative economics," *American economic review*, 2017, *107* (4), 967–1004.

_ , "Narrative economics," in "Narrative Economics," Princeton University Press, 2020.

Sundararaman, Dhanasekar, Shijing Si, Vivek Subramanian, Guoyin Wang, Devamanyu Hazarika, and Lawrence Carin, "Methods for Numeracy-Preserving Word Embeddings," in "Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)" 2020, pp. 4742–4753.

Swanson, Eric T, "Measuring the effects of Federal Reserve forward guidance and asset purchases on financial markets," Technical Report, National Bureau of Economic Research 2017.

Woodford, Michael, "Central Bank Communication and Policy Effectiveness," *Proceedings of the Federal Reserve Bank of Kansas City Economic Symposium at Jackson Hole*, 2005, pp. 399–474.

Supplementary Online Appendix

(Not for publication)

Deciphering Federal Reserve Communication via Text Analysis of Alternative FOMC Statements

Taeyoung Doh Dongho Song Shu-Kuei Yang

A Details about Text Analysis

A.1 Text Analysis using a Large Language Model

We briefly describe the procedure that we use to analyze text data using a large language model.[1] The raw text data is unstructured and can be represented as a string of characters, including letters, numbers, and symbols. We denote the total number of relevant characters as $|C|$. Consequently, each text is a string of characters with a variable length, as illustrated in the following example:

$$\text{Text}_t = [H, o, w, <>, a, r, e, <>, y, o, u, ?] \in |C|^{12}. \tag{A-1}$$

Because this data is unstructured, distinguishing different text data using a distance metric can be challenging. The embedding of text data by a large language model provides a representation of the original string data as a numeric vector, allowing us to define various text data features within a vector space. In the model, this embedding process involves two distinct steps: 1) tokenization, 2) neural network architectures.

A.1.1 Tokenization

Characters do not have linguistic or statistical meaning by themselves. A language model converts the sequence of characters into the sequence of tokens that are more interpretable linguistically or statistically. Language models transform the raw input text into the sequence of tokens by using tokenizers. One of popular tokenizers is PTB tokenizer, which obeys the

[1] Technical discussion in this section heavily draws on lectures notes on a large language model offered by the computer science department at Stanford University. For more details, see the above link and references therein.

English grammar and separates the contraction into two units. USE applies the PTB tokenizer to the raw input text. After the tokenization, the input data is transformed from the sequence of characters to the sequence of tokens, in which each token typically corresponds to a distinct vocabulary.

$$\phi : |C|^l \rightarrow |V|^L \,,\ \text{Text}_{v,t} = \phi(\text{Text}_t), \tag{A-2}$$

$$[H, o, w, <>, a, r, e, <>, y, o, u, ?] \rightarrow [How, <>, are, you, ?]. \tag{A-3}$$

The length of tokens in the sequence (L) might differ from the length of characters in the sequence (l) due to the pairing of characters during tokenization. Each token has its own embedding as a numeric vector. We can denote this token embedding as w. The result of tokenization is a sequence of token embeddings that represent the input text as $[w_1, \ldots, w_L]$. Each token is represented as a 64-dimensional numeric vector.

A.2 Neural Network Architecture

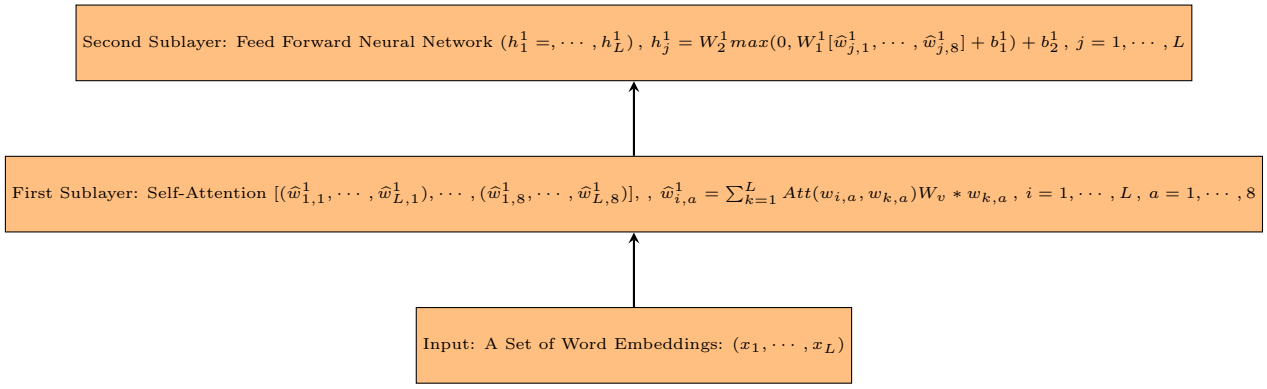
The USE has the transformer architecture consisting of six neutral network layers, each of which has two sublayers with eight self-attention heads. We describe tho original architecture and then explain how to fine tune it to obtain the paragraph level decomposition of similarity scoring across statements.

The first neural network in the USE is built by linking two sublayers as shown in Figure A-1 after taking a group of word embeddings that represents the source sentence as input. For the first layer, the same token embeddings are used for each attention head as input. In other words, $w^1_{i,a} = w_i \,, \forall a = 1, \cdots, 8$. The first layer generates the eight sequence of word embeddings $([\widetilde{w}^1_{1,1}, \cdots, \widetilde{w}^1_{L,1}], \cdots, [\widetilde{w}^1_{1,8}, \cdots, \widetilde{w}^1_{L,8}])$ as output and feeds this as input for the second layer. The actual USE architecture is slightly more complicated than presented below. It involves 1) positional embedding in which the order of any given word is also mapped into the embedding of that word,[2] 2) residual connection in which input bypasses attention and feed-forward neural network channels with a certain probability known as the dropout rate, 3) output from the layer is normalized to have mean zero and standard deviation of one.

The self attention channel can be best understood as looking up the dictionary value of all the words ($W_v * w_j \,, \forall j = 1, \cdots, L$) in the input text to match the query ($W_q * w_i$). The strength of match with the query word for each word in the look-up table is determined by the key ($W_k * w_j$). Here, attention weights for the a-th head are determined by how strong the key is with respect

[2]The position of each word embedding is used to generate since and cosine functions of different frequencies and these values are used the position embedding of the i-th word (P_i) and added to the input embedding w_i.

Figure A-1: First Neural Network Layer



to the query word. In the attention sublayer, every word embedding is linearly transformed to have the key, the query, and the value representation through W_k, W_q, W_v.

$$\text{(A-4)} \qquad \widehat{w}_{i,a} = \sum_{k=1}^{L} Att(w_{i,a}, w_{k,a}) W_v * w_{k,a} \ , \ Att(w_{i,a}, w_{k,a}) = \frac{e^{w'_{i,a} W'_k W_q w_{k,a}}}{\sum_{k=1}^{L} e^{w'_{i,a} W'_k W_q w_{k,al}}}.$$

Once word embeddings based on eight attention heads are obtained, USE concatenates these eight embeddings of each word into one big embedding and apply the feedforward neural network to this sequence of L embeddings. The final output from the second sublayer is the sequence of L embeddings.[3]

The second layer takes the output of the first layer as input and split the 512-dimensional vector representation (h_i^1) into eight 64-dimensional vector representations $([w_{i,1}^2, \cdots, w_{i,8}^2])$.

The entire USE algorithm works by vertically stacking six neural network layers which take the sentence embedding output in the previous layer as input and generate another sentence embedding as output. Figure A-2 describes the entire process.

To train parameters in the neural network architecture, we need to define the loss function that compares outcomes based on sentence embeddings from the USE with those based on human judgement. For example, if we define the relation between two texts as one of 3 classes (entail,contradict,neutral), we can apply the softmax classifier (f) to the difference between two embeddings. In this case, we can choose parameters in the neural network architecture to minimize the loss function that measures the distance between the machine-classified outcome $(f(U^i, U^j))$ and the one judged by humans $(f^{\text{human}}(\text{Text}_i, \text{Text}_j))$ where U^i is the 512-dimensional

[3]Each embedding representation has 512 dimension because we concatenate eight transformations of 64 dimensional original token embeddings.

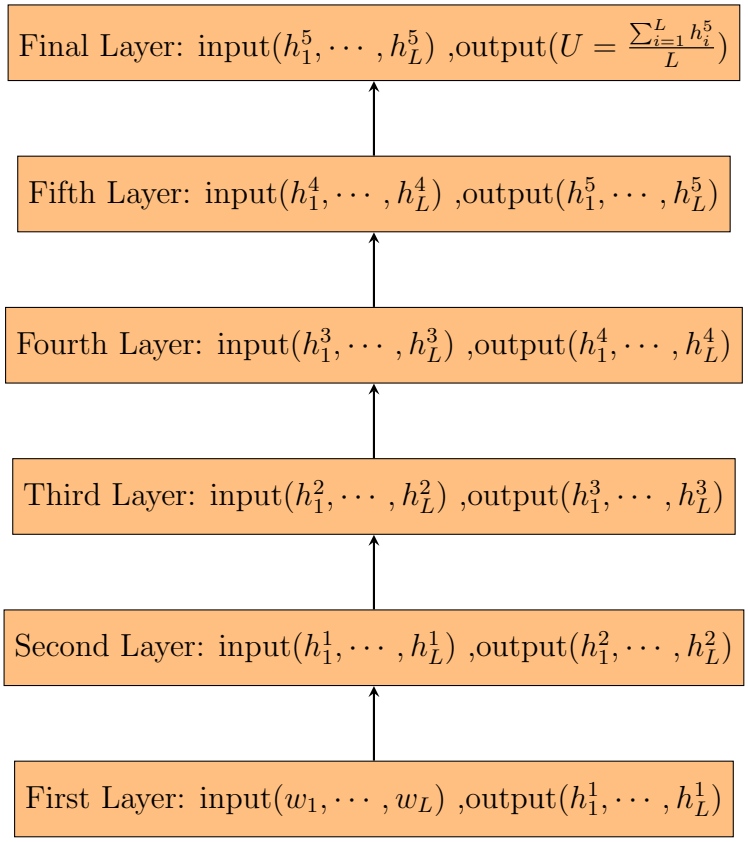


Figure A-2: Neural Network Architecture

USE representation of Text$_i$. In addition, two other natural language processing tasks are run to train the model.

- **Skip-thought task**: conditional on the center sentence, predict neighboring sentences (previous and next). The training dataset is from wikipedia articles.
- **Question-answer prediction**: predict the correct response for a given question among a list of correct answers and other randomly sampled answers. The training dataset is from web question-answer pages and discussion forums.
- **Natural language inference**: given a premise sentence and a hypothesis sentence, extract the relation between them. Let U_p and U_h be the sentence embeddings of the premise and the hypothesis, respectively. A fully-connect layer and and a 3-way softmax classifier are applied for the concatenated input of (U_p, U_h, $|U_p - U_h|$, and $U_p - U_h$). The three-way classifier predicts if the premise entails, contradicts, or is neutral to the hypothesis. The training dataset is the Stanford Natural Language Inference (SNLI) corpus.

A.3 Paragraph Level Decomposition of the USE Representation

In some cases, paragraph-by-paragraph comparison may provide more interpretable results. For instance, we may be interested in which paragraph drives the similarity score between different statements. For this, we obtain paragraph level USE representations and approximate the statement level USE representation by a weighted average of paragraph level USE representations.

Denote the USE representation of the released FOMC statement at time t by S_t^R. Similarly, S_t^i , $(i = A, B, C, D)$ denotes the USE representation of alternative statements. The USE representation of the j-th paragraph of the FOMC statement at time t is $P_{j,t}^i$. To calculate $P_{j,t}^i$, we run the USE algorithm for each paragraph j. The idea is to construct $\sum_k w_k P_{k,t}^i$ that can mimic S_t^i best in terms of minimizing the squared difference between two representations of the FOMC statement at time t.

- **Step 1: Paragraph Padding** Some statements are longer than others, meaning that the corpus of FOMC statements has an unequal length depending on the statement. An easy way to fix this is to pad a shorter statement with empty paragraph encodings. Suppose that n_{max} is the maximum number of paragraph of any given FOMC statement from the entire corpus of our dataset including both released statements and alternative statements. Then, we can extract the following array of the paragraph USE representation of the FOMC statement.

$$P_t^R = [P_{1,t}^R, \cdots, P_{n_{max},t}^R]. \tag{A-5}$$

  If the number of paragraphs in the statement at time t ($n_{R,t}$) is smaller than n_{max}, we add $(n_{max} - n_{R,t})$ zero vectors of 512 dimensions. The purpose of this operation is to make the USE representation of any FOMC statement have the same number of the USE representations at the paragraph level.

- **Step 2: Approximate the Statement Level USE Representation by a Weighted Average of Paragraph Level USE Representations**

  The goal is to select weights $(w_j\ ,\ j = (1, \cdots, n_{max})$ that can mimic this statement-level USE representation using paragraph-level USE representations. We consider the following squared loss:

$$\sum_{i \in R,A,B,C,D} \sum_t (S_t^i - \sum_j w_j P_{j,t}^i)^T (S_t^i - \sum_j w_j P_{j,t}^i). \tag{A-6}$$

We can put the non-negativity and unit-sum constraints on w_j such that $w_j >= 0$, $\sum_j w_j = 1$. Once we find the solution for weights, we can mimic P_t^i by $\sum_j w_j P_{j,t}^i$. But the numerical optimization routine might be non-convex when you put the constraints directly. So we may consider the following transformation of w_j to make the problem an unconstrained minimization problem:

$$w_j = \frac{e^{\alpha_j}}{\sum_{k=1}^{n_{max}} e^{\alpha_k}}, \tag{A-7}$$

where α_j is an unconstrained parameter. Notice that w_j still satisfies the constraints but we are minimizing the loss function with respect to $(\alpha_1, \cdots, \alpha_{n_{max}})$.

- **Step 3: Decomposing the Similarity Scoring**

  For the unit-vector, the cosine similarity is simply the inner product. So we can renormalize the USE representation to have a unit length. In that case, we have the following nice decomposition of the similarity scoring between texts.

$$Sim(P_t^i, P_t^j) \propto Sim(\sum_{k=1}^{n_{max}} w_k P_{k,t}^i, \sum_{k=1}^{n_{max}} w_k P_{k,t}^j) = \sum_k \sum_{k'} w_k w_{k'} Sim(P_{k,t}^i, P_{k',t}^j). \tag{A-8}$$

A.4 Details of Fine-tuning

As explained in the text, we add an additional layer to the USE representation of the text to train the final embedding output to recognize numeric properties better. We consider a fully connected feed-forward network with a rectified linear unit as an activation function. For the original USE representation of a FOMC statement $U_{FOMC} = [U_1, \cdots, U_{512}]$, our additional layer performs the following transformation:

$$f(U_{FOMC}) = [\max(W_1' U_{FOMC} + b_1, 0), \cdots, \max(W_{512}' U_{FOMC} + b_{512}, 0)]. \tag{A-9}$$

Let's stack parameters governing this transformation by $\vartheta = [W_1, \cdots, W_{512}, b]$ where $b = [b_1, \cdots, b_{512}]$. As described in the text, we generate two separate training datasets to optimize ϑ in order to minimize loss functions set out in equation (2) and (3).

B Prediction Regression

Consider the following linear regression model:

$$y_t = z'_{t-\Delta}\delta + e_t, \quad e_t \sim \mathcal{N}(0, \sigma^2), \tag{A-10}$$

where $y_t \in \mathbb{R}$ and $z_{t-\Delta} \in \mathbb{R}^k$ is the full predictor vector observed at time $t-\Delta$. In the presence of missing observations in $z_{t-\Delta}$, we define a row selection matrix $M_{t-\Delta} \in \mathbb{R}^{k_{t-\Delta} \times k}$, with $k_{t-\Delta} < k$, that extracts only the observed components of $z_{t-\Delta}$. In this case, we can re-express (A-10) as:

$$y_t = x'_{t-\Delta}\delta + e_t, \quad e_t \sim \mathcal{N}(0, \sigma^2), \tag{A-11}$$

where $x'_{t-\Delta} \equiv z'_{t-\Delta}M'_{t-\Delta}M_{t-\Delta}$, and $M'_{t-\Delta}M_{t-\Delta} \in \mathbb{R}^{k\times k}$ is a diagonal matrix that zeroes out unobserved elements of $z_{t-\Delta}$.

B.1 Posterior updating rules

This formulation of (A-11) ensures that only the observed entries contribute to posterior learning.

Prior distributions. We assume the following prior distributions:

$$\delta \mid \sigma^2 \sim N(\delta_{t-1|t-1}, \sigma^2 V_{t-1|t-1}), \quad \sigma^2 \sim \text{IG}\left(\frac{\nu_{t-1}}{2}, \frac{\nu_{t-1}s^2_{t-1}}{2}\right). \tag{A-12}$$

Given a new observation $(y_t, x_{t-\Delta})$, the likelihood function is:

$$p(y_t \mid \delta, \sigma^2, x_{t-\Delta}) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{(y_t - x'_{t-\Delta}\delta)^2}{2\sigma^2}\right).$$

Posterior distributions. The posterior distribution of σ^2 follows an Inverse-Gamma

$$\sigma^2 \mid y_t, x_{t-\Delta} \sim \text{IG}\left(\frac{\nu_t}{2}, \frac{\nu_t s^2_t}{2}\right), \tag{A-13}$$

with the update as follows:

$$\nu_t = \nu_{t-1} + 1, \quad \nu_t s^2_t = \nu_{t-1}s^2_{t-1} + \frac{(y_t - x'_{t-\Delta}\delta_{t-1|t-1})^2}{1 + x'_{t-\Delta}V_{t-1|t-1}x_{t-\Delta}}.$$

Conditionally on σ^2, the posterior distribution of δ remains normal:

$$\delta \mid y_t, x_{t-\Delta}, \sigma^2 \sim N(\delta_{t|t}, \sigma^2 V_{t|t}), \tag{A-14}$$

where the posterior mean and variance update as follows:

$$\delta_{t|t} = V_{t|t}\left(V_{t-1|t-1}^{-1}\delta_{t-1|t-1} + x_{t-\Delta}y_t\right), \quad V_{t|t} = \left(V_{t-1|t-1}^{-1} + x_{t-\Delta}x_{t-\Delta}'\right)^{-1}.$$

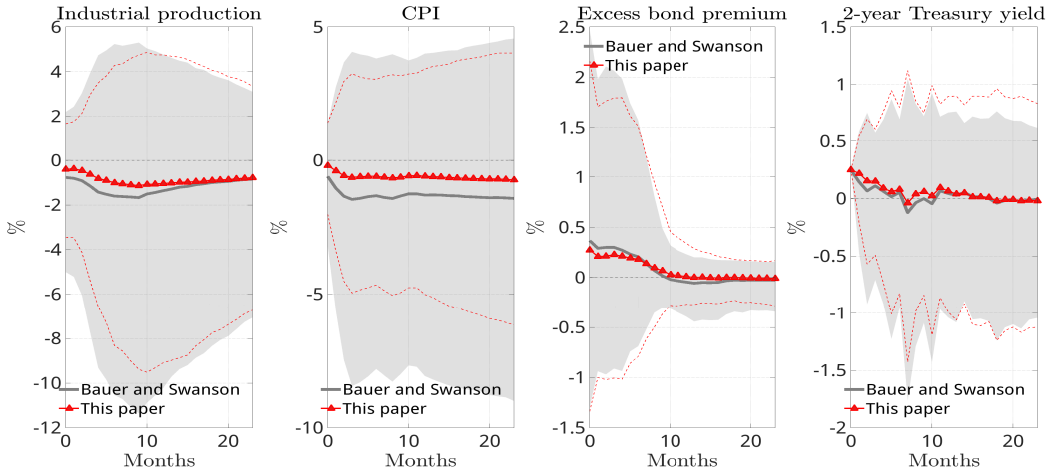
C Residual-based Moving Block Bootstrap (MBB) Confidence Intervals for SVAR IRFs

Confidence intervals for impulse response functions in Figure 3 are computed by the wild bootstrap method used in Bauer and Swanson (2023a). Jentsch and Lunsford (2022) show that the method is not asymptotically valid and can result in misleading confidence intervals and coverage rates for time series data with serial dependence. As an alternative, they propose a residual-based moving block bootstrap (MBB) method that can address these issues. We recalculate 90% and 68% confidence intervals for SVAR impulse-responses in the main text using the residual-based moving block method in Figure A-3.

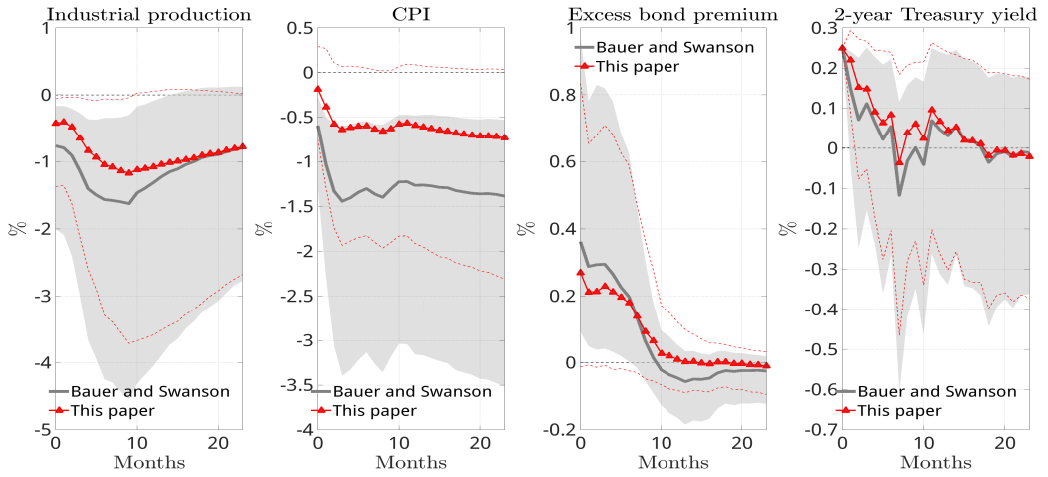
Compared to 90% confidence intervals in Figure 3 based on the wild bootstrapping method, we observe wider bands in general. Both monetary policy shocks identified by the method in Bauer and Swanson (2023a) and our method lose statistical significance in 90% confidence bands when we use MBB. Nonetheless, a contractionary monetary policy shock reduces industrial production and CPI while increasing the excess bond premium in a statistically significant way if we focus on the 68% confidence interval in Figure A-3. Point estimates based on the median IRFs are robust to the use of different bootstrapping methods.

Figure A-3: SVAR IRFs with MBB confidence intervals

Panel A. 90% Confidence Interval



Panel B. 68% Confidence Interval



Notes: Following Bauer and Swanson (2023b), we estimate a SVAR with a 12-month lag using the following variables: log industrial production, log consumer price index, excess bond premium Gilchrist and Zakrajšek (2012), and the two-year Treasury bond yield. To instrument our analysis, we employ the orthogonalized monetary policy surprise measure derived from the residuals of a regression on six predictors explained in Table 4 observed before the FOMC announcement. While we estimate the VAR coefficients using a longer sample from 1973:M1 to 2019:M12, the orthogonalized high-frequency monetary policy surprises are available only from 2004:M3 to 2016:M12. To facilitate comparison, we present the impulse responses obtained by utilizing the orthogonalized high-frequency monetary policy surprise instrument employed in Bauer and Swanson (2023b). Confidence intervals are calculated based on the residual-based moving block bootstrapping method in Jentsch and Lunsford (2022).